Cloudra: Geometric Reasoning and Scientific Validation for Autonomous Reaction Mechanism Exploration

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Abstract

Autonomous reaction-mechanism exploration requires proposing reaction steps, preparing three-dimensional structures, and determining which reactant and product a computed saddle point connects. Large language model (LLM) agents can misjudge spatial relationships and confuse a computed structure with the species or step they intended to study.

We present Cloudra, a multi-agent framework for exploring reaction mechanisms. A Delivery DAG represents the species and elementary steps being studied. Under intelligent restraint, deterministic algorithms match atoms between reactants and products, measure bond lengths and dihedral angles, rotate and translate molecular fragments, and assemble candidate transition-state geometries. Frequency calculations distinguish minima from saddle points, while IRC endpoint calculations identify the species connected by each saddle. These operations let the agents test proposed steps, compare competing pathways, and revise the mechanism.

CloudraBench evaluates the framework's end-to-end mechanism-calculation capability across 50 tasks covering elementary reactions, multi-step mechanisms, stereochemical competition, and computational method limitations. Across 200 runs under four workflow configurations, Cloudra achieves a mean composite score of 0.793, compared with 0.581 for the same-model Standard baseline. On Hard-tier tasks, the mean score increases from 0.689 to 0.958. The self-consistency component reaches its maximum score in 24 of the 40 non-Impossible tasks, compared with seven under Standard. Behavioral analyses examine how the agents use bond and dihedral measurements, optimized structures, and IRC endpoints to revise local mechanistic assignments.

1 Introduction

1.1 Physical requirements for autonomous reaction mechanism research

Scientific validation of an organic reaction mechanism requires checking the molecular geometries, reaction pathways, and quantum-chemical results assigned to each species and elementary step. Normal termination of a geometry optimization (exit 0) records completion of the program. Identifying a stable intermediate requires frequency analysis to confirm a minimum. Assigning a first-order saddle to a reaction step requires IRC calculations that identify the species connected on both sides. An intrinsic reaction coordinate (IRC) that reaches an unintended dissociation product or stereoisomer also changes the mechanistic assignment of the associated energy calculation. These checks are particularly important when a study includes several elementary steps or competing stereochemical pathways.[1, 2]

Computational chemistry agents automate several parts of this work. El Agente Q [3], Aitomia [4], and Gráfico [5] orchestrate quantum-chemical calculations and typed workflows. Estructural [6] and ARCHE [7] provide tools for molecular fragment editing, atom mapping, and transition-state geometry preparation; Quntur [8] addresses reaction kinetics and transition-state research. Extending these capabilities to autonomous mechanism exploration requires reliable geometric operations and physical interpretation of the species and pathways obtained.

Aligning stereochemical orientations, assembling reactant fragments, and preparing pathway endpoints require three-dimensional geometric reasoning. Language models can make errors when they infer spatial relationships or directly manipulate Cartesian coordinates. Quntur [8], for example, reports the need for human assistance with endpoint alignment in complex pathway and transition-state preparation. In Cloudra, an agent can ask the program to rotate the atoms on one side of a bond, measure the resulting dihedral angle, and compare the new geometry with another transition-state guess.

A proposed intermediate must optimize to a minimum. A candidate transition state must be a first-order saddle whose paths lead to the intended reactant and product. If an IRC reaches another product or stereoisomer, the saddle belongs to a different reaction from the one initially proposed. An autonomous system must use these results to reassign the step or search for another saddle.

1.2 Cloudra: autonomous mechanism exploration through chemical objects

We introduce Cloudra, a multi-agent system for exploring reaction mechanisms involving small organic molecules. Language models propose candidate pathways, choose calculations, and revise hypotheses from the returned structures and energies. The software maps atoms, moves molecular fragments, and runs geometry optimizations, frequency calculations, and IRCs. The results identify minima and saddle points and determine which species each saddle connects.

Figure 1 summarizes a structured literature sample of 85 chemical AI agent systems: 39 concern materials or surfaces, 23 concern finite molecules or complexes, 12 concern soft matter or biomolecules, and 11 cover multiple categories of objects. Figures 2 and 3 show the documented tasks in the finite-molecule/complex category and compare Cloudra's capabilities with those of the existing systems. The accompanying literature materials contain the complete matrix and the sources for each entry.

Following intelligent restraint, Cloudra assigns coordinate calculations to deterministic routines. MCS matching selects molecular fragments; atom mapping matches reactant and product atoms and identifies atom pairs whose bonds change. The program measures bond lengths, bond angles, dihedral angles, and stereochemistry, and rotates or translates selected atoms to prepare candidate structures.

The Delivery DAG represents chemical species as nodes

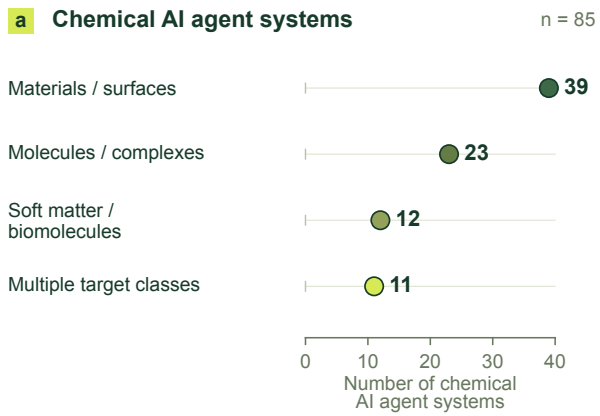


Figure 1. Physical targets in a structured literature sample of 85 chemical AI agent systems: 39 materials/surface, 23 finite-molecule/complex, 12 soft-matter/biomolecular, and 11 cross-class systems.

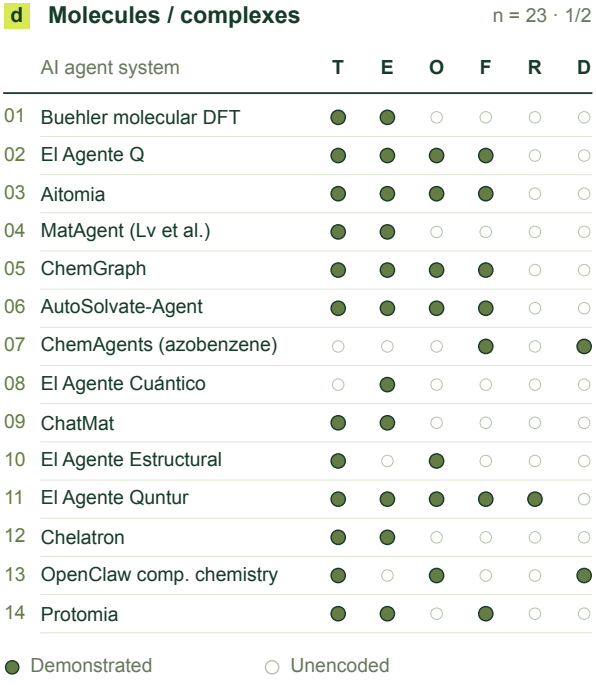


Figure 2. Documented tasks in the finite-molecule/complex class (panel 1 of 2; 23 chemical AI agent systems in total). T: structure construction/editing; E: static energy/properties; O: geometry optimization; F: vibrations/thermochemistry; R: explicit pathways/TS; D: molecular/quantum dynamics. Filled cells denote positive evidence; blanks leave capability coverage unspecified.

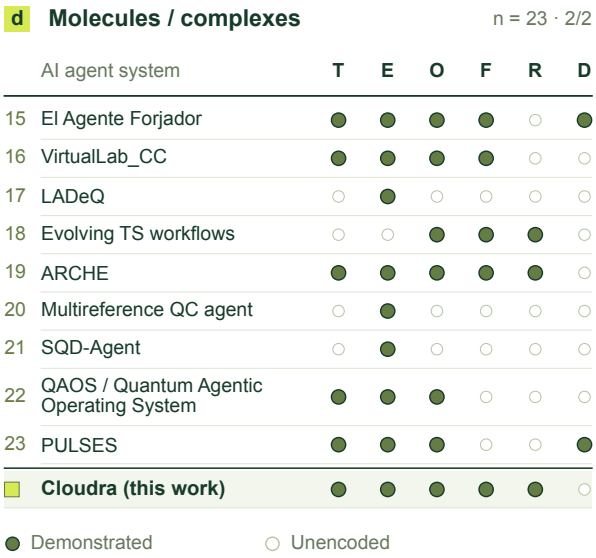


Figure 3. Chemical AI agent systems in the finite-molecule/complex class, continued (panel 2 of 2), using the codes in Figure 2. Cloudra (this work) is benchmarked at the bottom. Complete matrices for all 85 chemical AI agent systems and source-level qualifications accompany the draft.

and elementary reaction steps as edges. Nodes link to optimized geometries, energies, and frequencies; edges link to transition states and their IRC paths. A transition state must have exactly one imaginary frequency. The program checks that its frequency calculation and both IRC directions refer to the same saddle, and that optimizations from the IRC terminal frames reach the declared reactant and product species. The Supervisor reviews the resulting mechanism against the task and identifies calculations still needed and steps requiring further investigation.

1.3 Contributions

Cloudra combines a DAG of species and elementary steps with algorithms for atom mapping, geometry construction, and stationary-point and pathway analysis. Atom mappings keep reactant and product atoms in a common order. Fragment operations generate candidate geometries, and frequency and IRC calculations determine which stationary points were found and which species they connect. The agents use these results to compare proposed steps, revise the mechanism, and select further calculations.

The geometric procedures include MCS-based fragment selection, graph traversal to select the atoms that move, and transition-state fragment assembly. They allow the agents to rotate selected groups, adjust fragment positions, measure the resulting structures, and prepare new guesses after an unsuccessful transition-state search.

We evaluate the framework's end-to-end mechanism-calculation capability using CloudraBench, which measures graph coverage, physical self-consistency, and task-defined target accuracy from native computational outputs. Its 50 tasks cover elementary and multistep reactions, stereochemistry, and the applicability limits of the allowed methods. The experiments comprise 48 independent runs with three repeats on four tasks and a completed study of 200 runs under four workflow configurations across all 50 tasks. Standard uses the same model backbone as the primary comparison;

Record and Brief-errors examine evidence organization and diagnostic feedback.

2 System Design and Methods

2.1 System Overview and Division of Responsibilities

Cloudra investigates reaction mechanisms by constructing candidate molecular configurations, locating and characterizing stationary points, tracing elementary reaction paths, and comparing competing channels under the specified reaction conditions and computational requirements. The Delivery DAG represents species as nodes and elementary steps as edges. Atom mapping identifies corresponding atoms in reactant and product structures; geometric operators rotate, translate, and assemble molecular fragments. Frequency and IRC calculations test the stationary points and paths assigned to each species or step.

The Coordinator proposes mechanistic hypotheses and candidate elementary steps, and selects which branches to investigate. Computational agents prepare three-dimensional structures with the geometric operators and submit quantum-chemical calculations for their assigned steps. If a geometry diverges, a saddle-point optimization fails to converge, or a path bifurcates, the computational agent examines the structures and calculation diagnostics to decide whether to continue the step. If the step appears blocked, it reports the diagnosis to the Coordinator, which revises the hypothesis or explores another branch. The Supervisor reviews the species, steps, and calculation results in the DAG against the task requirements. Its review notes are stored separately from the facts parsed from calculation outputs.

2.1.1 Asynchronous Computation Scheduling and Resource Management

The central calculation ledger (`ComputeDB`) records each job's state. Separate capacity pools limit concurrent calculations on local workers, remote SSH clusters, and Slurm queues. Before dispatch, the scheduler computes a SHA-256 digest from the Cartesian coordinates, spin, charge, and level of theory. The scheduler reuses a matching active job or cached result. Operating-system exclusive file locks (`flock`) track running processes and concurrency; quantum-chemistry processes holding active locks remain running during monitor exceptions. When a calculation ends, a scheduler notification triggers asynchronous output reading by the computational agent.

2.2 Molecular Representations and Geometric Operators

A molecular input consists of an attributed graph $H = (A, B)$ and Cartesian coordinates $\mathbf{X} \in \mathbb{R}^{n\times 3}$. Here A is the atom set, B is the bond set, and each row of $\mathbf{X}$ gives one atom position. The graph stores atom and bond attributes. Geometric operators change the coordinates and return measurements of the resulting structure. Replacing atoms or functional groups also changes H.

2.2.1 Atom Correspondence and Reaction Centers

Atom attributes retain element, isotope, formal charge, hydrogen count, radical-electron count, and declared stereochemistry; bond attributes retain order and stereochemical labels. Existing coordinates are retained. ETKDGv3 [9–11] embeds the remaining components, which are then assembled if necessary. Complete, unique atom-map labels on both endpoints define the correspondence directly and must agree in element and isotope. Otherwise, let A_{P} and A_{R} be the product and reactant heavy-atom sets. Candidate bijections $\pi : A_{\mathrm{P}} \to A_{\mathrm{R}}$ satisfy $z_i^{\mathrm{P}} = z_{\pi(i)}^{\mathrm{R}}$ and $\iota_i^{\mathrm{P}} = \iota_{\pi(i)}^{\mathrm{R}}$, where z is atomic number and ι is isotope label. Each candidate is scored by

$$J(\pi) = 2(N_{\mathrm{f}} + N_{\mathrm{b}}) + N_{\mathrm{o}} + 4\Delta_H + 8\Delta_q + 8\Delta_r + 2N_{\mathrm{s}}. \tag{2.1}$$

Let B_{P}^{π} denote product bonds expressed in reactant atom indices. Bond formation and cleavage are the set differences $B_{\mathrm{P}}^{\pi} \setminus B_{\mathrm{R}}$ and $B_{\mathrm{R}} \setminus B_{\mathrm{P}}^{\pi}$, with cardinalities N_{f} and N_{b}. The count N_{o} includes pairs in $B_{\mathrm{P}}^{\pi} \cap B_{\mathrm{R}}$ whose bond orders differ. For $a \in \{H, q, r\}$, the mismatch is $\Delta_a = \sum_{i\in A_{\mathrm{P}}} |a_i^{\mathrm{P}} - a_{\pi(i)}^{\mathrm{R}}|$, where H, q, and r are the atom hydrogen count, formal charge, and radical-electron count. N_{s} counts stereochemical mismatches. Exactly one declared tetrahedral label contributes one mismatch, increasing J by 2. When both corresponding atoms carry tetrahedral labels, the comparison requires four corresponding neighbors, including a single hydrogen where applicable; it corrects for neighbor-order parity before comparing the labels. For two-sided labels, only comparable environments enter mismatch scoring.

For element–isotope classes of sizes n_g, the search space contains $N_\Pi = \prod_g n_g!$ bijections. All are enumerated when $N_\Pi \le 10^5$. Above this limit, MCS seeds use element matching, unrestricted bond-order matching, ring-to-ring matching, and complete rings [11]. The MCS timeout is 10 s, and each endpoint contributes at most 64 substructure matches. Unmatched atoms are paired within element–isotope classes after sorting by hydrogen count, degree, and input index. Candidates with the same formed, broken, and order-changed bonds and the same score are merged, keeping the lexicographically smallest mapping for each group. Representatives are sorted by score and then atom-map sequence; the first 16 form the heavy-atom candidate set C. For nonempty C, define optimal score $J_{\min} = \min_{\pi\in C} J(\pi)$ and tolerance set $\mathcal{N} = \{\pi \in C : J(\pi) \le J_{\min} + 2\}$. An unlabeled mapping is reported as unique when $|\mathcal{N}| = 1$, ambiguous when $|\mathcal{N}| > 1$, and inconclusive when $C = \emptyset$.

Each retained heavy-atom mapping is expanded to all atoms using matches compatible with both endpoint graphs. Compatible explicit hydrogen labels are preserved first; remaining hydrogens are paired by isotope on corresponding heavy atoms. For remaining isotope groups of sizes h_g, the remaining $N_H = \prod_g h_g!$ permutations are enumerated when $N_H < 257$. An expansion with $N_H \ge 257$ requires compatible explicit hydrogen labels that resolve the permutation space; unresolved expansions are reported as inconclusive. Complete maps are deduplicated, ranked by score and atom-map sequence, and evaluated with the same +2 ambiguity criterion. The selected all-atom bijection reorders the product into reactant atom order. Formed and broken bonds identify the atom pairs monitored during path interpolation; bond-order changes are recorded with the transformed graph.

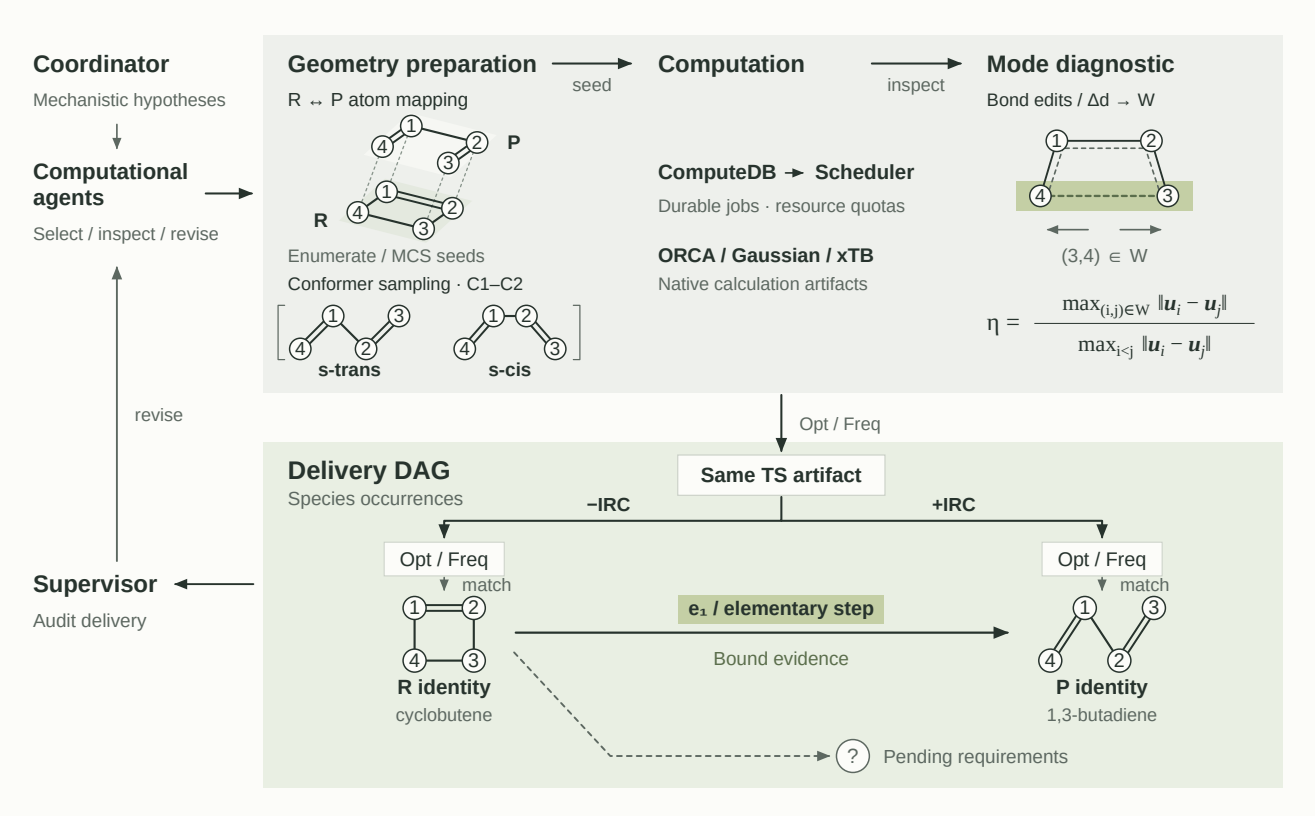


Figure 4. Cloudra's mechanism exploration workflow. Atom mapping and conformer sampling prepare reactant and product structures; ComputeDB and the scheduler manage the calculations. Mode diagnostics measure motion on monitored atom pairs. Bidirectional IRCs start from the same transition state, and their outer frames undergo endpoint optimization and frequency analysis. The resulting structures are compared with the species declared in the Delivery DAG. The Supervisor reviews the graph and identifies calculations to complete and assignments to resolve.

2.2.2 Structural Measurements and Rigid Body Operations

The system calculates bond lengths, bond angles, dihedral angles, ring conformations, and close contacts from the coordinates. For coordinates $\mathbf{x}_i$, the interatomic distance is $d_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$; the bond angle at vertex j is obtained from the normalized dot product of the two bond vectors. Distances are compared with reference bond lengths and van der Waals contact scales to flag geometric anomalies. For coordinate-only inputs, connectivity is determined at entry. Local coordinate transformations retain this bond graph and repeat the measurements before and after moving atoms.

Default fragment selection on either side of an acyclic bond uses graph connectivity: the specified bond $a-b$ is temporarily removed, and the connected component containing b, denoted $\mathcal{F}_b$, is selected. If the endpoints remain connected after bond removal, movement within a ring framework requires an explicit atom group. Let $\mathcal{F}$ denote the explicitly specified moving atom group or the component $\mathcal{F}_b$ obtained by bond removal. To rotate the selected fragment about an axis, Rodrigues' formula is applied to every $i \in \mathcal{F}$:

$$\begin{aligned} \mathbf{x}_i' &= \mathbf{a} + \mathbf{R}(\hat{\mathbf{u}}, \theta)(\mathbf{x}_i - \mathbf{a}), \\ \mathbf{R} &= \cos\theta\, \mathbf{I} + (1 - \cos\theta)\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}} \\ &\quad + \sin\theta\, [\hat{\mathbf{u}}]_{\times}. \end{aligned} \tag{2.2}$$

Here, $\mathbf{a}$ is a point on the axis, $\hat{\mathbf{u}}$ its unit direction, θ the rotation angle, $\mathbf{I}$ the identity matrix, and $[\hat{\mathbf{u}}]_{\times}\mathbf{v} = \hat{\mathbf{u}} \times \mathbf{v}$. The operation fixes the coordinates of atoms outside the selected set. Stretching bond $a - b$ translates $\mathcal{F}_b$ as a whole:

$$\mathbf{x}_i' = \mathbf{x}_i + (L - d_{ab})\frac{\mathbf{x}_b - \mathbf{x}_a}{d_{ab}}, \qquad i \in \mathcal{F}_b. \tag{2.3}$$

Here L is the target bond length; coordinates outside $\mathcal{F}_b$ remain fixed. Chirality inversion rotates two separable branches around their center to exchange branch directions. Group substitution removes the specified-side fragment, docks the new group along the original attachment-bond direction, and updates the graph. To compare structures before and after optimization, the system differences distance-perceived connectivity sets under the same element sequence and classifies gained or lost contacts as intra- or interfragment changes.

2.2.3 Branch Crowding and Reactant Preorganization

After checking hybridization, ring structure, and substituent graph inequivalence, the system calculates crowding around each candidate axis from the atoms in its substituent branches:

$$B_s = \sum_{k \in T_s} c_k r_k \rho^{\ell_k}, \qquad \rho = 0.5, \quad 0 \le \ell_k \le 5. \tag{2.4}$$

Here r_k represents the van der Waals radius of atom k. T_s is generated by a last-in-first-out depth-first traversal starting from the substituent root at depth zero, with the adjacent axis-end atom marked visited. On popping an unvisited atom, the algorithm records its traversal depth ℓ_k and pushes its unvisited neighbors if $\ell_k < 5$. The weighting coefficient

is $c_k = 1.5$ for a root of graph degree at least 3, and $c_k = 1$ otherwise.

The descriptor uses standard van der Waals radii in Å: H 1.20, B 1.92, C 1.70, N 1.55, O 1.52, F 1.47, P 1.80, S 1.80, Cl 1.75, Br 1.85, and I 1.98 (1.70 fallback for unlisted elements). The four branch values are rounded to three decimal places and summed to obtain the axis score B_{axis}; $B_{\text{axis}} \geq 9.0$ flags the axis as congested for subsequent conformational inspection.

During multicomponent preorganization, the primary fragment is fixed at the origin. Remaining fragments undergo orientation sampling on the unit sphere. With K samples, direction s is

$$\begin{aligned} \hat{\mathbf{u}}_s &= (a_s \cos s\gamma,\ y_s,\ a_s \sin s\gamma), \\ a_s &= \sqrt{1 - y_s^2}, \qquad y_s = 1 - \frac{2s}{K-1}, \\ \gamma &= \pi(3 - \sqrt{5}). \end{aligned} \tag{2.5}$$

Here $\gamma = \pi(3 - \sqrt{5}) \approx 2.3999$ rad ($137.5°$) represents the Golden Angle, generating a Fibonacci spherical spiral grid for uniform distribution of directions. The axial twist about $\hat{\mathbf{u}}_s$ is $2\pi s/K$ (default $K = 32$). Translation along the interfragment vector resolves steric overlap. For a set W of guided atom pairs with target distance windows $[l_{ij}, u_{ij}]$, placement $\mathbf{X}$ is scored by

$$\begin{aligned} S(\mathbf{X}) = \max_{(i,j)\in W} \{0,\ & l_{ij} - d_{ij}(\mathbf{X}), \\ & d_{ij}(\mathbf{X}) - u_{ij}\}. \end{aligned} \tag{2.6}$$

Configurations with low $S(\mathbf{X})$ undergo distance-restrained MMFF94 force-field optimization[12] to bring the specified atom pairs into their target distance windows while relaxing nonbonded contacts. Candidates are retained only if the force-field optimization converges, the distances fall within their windows, stereochemistry is preserved, and fragment contacts satisfy $d_{i^*j^*} \leq 1.7(r_{i^*} + r_{j^*})$. The retained configurations are ranked by unconstrained single-point energy and clustered with a heavy-atom Kabsch RMSD threshold of 0.5 Å to provide starting structures.

2.2.4 Endpoint Alignment and IDPP Interpolation

After reordering the product atoms to match the reactant, each endpoint is centered. Let $\mathbf{P}$ and $\mathbf{R}$ be the centered product and reactant coordinate matrices. Kabsch alignment[13] solves

$$\begin{aligned} \min_{\mathbf{Q}^\mathsf{T}\mathbf{Q}=\mathbf{I},\det\mathbf{Q}=1} \|\mathbf{PQ} - \mathbf{R}\|_\mathrm{F}^2, \\ \mathbf{P}^\mathsf{T}\mathbf{R} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\mathsf{T}, \\ \mathbf{Q} = \mathbf{U}\,\mathrm{diag}\big(1, 1, \det(\mathbf{U}\mathbf{V}^\mathsf{T})\big)\mathbf{V}^\mathsf{T}. \end{aligned} \tag{2.7}$$

The resulting matrix $\mathbf{Q}$ is the proper rigid-body rotation matrix. The determinant correction factor $\det(\mathbf{U}\mathbf{V}^\mathsf{T})$ enforces $\det\mathbf{Q} = +1$, ruling out improper reflections and preserving chiral configuration.

For M images including endpoints, define $\lambda_k = k/(M-1)$ and initialize $\mathbf{X}_k = (1-\lambda_k)\mathbf{R} + \lambda_k\mathbf{PQ}$. The image-dependent pair potential (IDPP) method [14] interpolates each image's target pair distances between the endpoints and optimizes its coordinates with the potential

$$\begin{aligned} d^*_{ij,k} &= (1-\lambda_k)d^\mathsf{R}_{ij} + \lambda_k d^\mathsf{P}_{ij}, \\ V_k &= \sum_{i<j} \frac{(d_{ij,k} - d^*_{ij,k})^2}{d^4_{ij,k}}. \end{aligned} \tag{2.8}$$

The inverse-fourth-power weighting $d^{-4}_{ij,k}$ imparts stiffness to short covalent bonds, prioritizing the preservation of basic covalent distances while allowing long-range nonbonded contacts to relax. Standard mode couples intermediate images with improved-tangent nudged elastic band (NEB) springs and optimizes with MDMin, evaluating convergence by the maximum force residual across movable atoms.[15, 16] Lightweight mode evaluates target distances with an objective denominator $\max(d^*_{ij,k}, 0.1\,\text{Å})^4$ and relaxes intermediate images independently using Adam optimization.[17]

Active monitoring pairs W incorporate bond-state changes, pairs bonded at both endpoints with $|\Delta d| \geq 0.15$ Å, and the five atom pairs exhibiting the largest endpoint distance variations. Within the monitoring set W, the primary reaction coordinate is selected by $p^* \in \mathrm{argmax}_{p\in W} |d^\mathsf{P}_p - d^\mathsf{R}_p|$, setting $m = (d^\mathsf{R}_{p^*} + d^\mathsf{P}_{p^*})/2$. The initial transition-state guess is extracted from the first image k satisfying $(d_{p^*,k} - m)(d^\mathsf{R}_{p^*} - m) \leq 0$, or the image minimizing $|d_{p^*,k} - m|$. The selected configuration serves as the initial seed for saddle-point optimization.

2.3 Chemical Species and Reaction Steps in the Delivery DAG

2.3.1 Species Occurrences and Required Calculations

The Delivery DAG is $G = (V, E)$. Each vertex $v \in V$ represents a species at one position in the reaction sequence, and each directed edge $e = (v_i, v_j) \in E$ represents an elementary step. The mapping $\kappa : V \to \mathcal{S}$ gives the canonical chemical identity of each species occurrence. An initial and a regenerated catalyst can therefore occupy separate nodes while sharing the same chemical identity. Species nodes record stationary-point geometries, frequencies, and absolute energies; step edges record transition-state geometries, imaginary-mode displacement vectors, and reaction paths.

For each species or step, K lists the required calculation stages. Standard elementary steps require the sequence $K = \{\text{ts_opt}, \text{frequency}, \text{short_irc}\}$, with thermochemical corrections added when required by the task rubric. Calculation outputs are archived as immutable records identified by content digests. Each parsed result is recorded as $f = (c, o, r, d)$: c names the check, $o \in \{+, -, ?\}$ records support, contradiction, or uncertainty, r identifies the source calculation output, and d contains the measured values.

For a species or step t, let F_t contain its parsed results, R its required property checks, and O its optional checks. The physical validation criterion is

$$\begin{aligned} S_t = \mathbf{1}[R \neq \emptyset] \\ \wedge \bigwedge_{c\in R} \mathbf{1}[\exists f \in F_t : f.c = c,\ f.o = +] \\ \wedge \mathbf{1}[\nexists f \in F_t : f.c \in R \cup O,\ f.o = -]. \end{aligned} \tag{2.9}$$

When $R \neq \emptyset$, each required check must have a supporting result, and every current result for a check in $R \cup O$

must indicate support or uncertainty. When $R = \emptyset$, only the calculation-stage requirements are checked.

2.3.2 Curvature and Reaction Mode Diagnostics

Stationary-point curvature is evaluated from the mass-weighted Hessian:

$$\begin{aligned} \mathbf{H}_{\text{mw}} &= \mathbf{M}^{-1/2} \nabla_{\mathbf{x}}^2 U \, \mathbf{M}^{-1/2}, \\ \mathbf{H}_{\text{mw}} \mathbf{v}_k &= \lambda_k \mathbf{v}_k. \end{aligned} \tag{2.10}$$

In the vibrational subspace orthogonal to translational and rotational degrees of freedom, negative eigenvalues $\lambda_k < 0$ correspond to imaginary harmonic frequencies $\nu_k < 0$. The number of imaginary frequencies is $n_{\text{imag}} = \sum_k \mathbf{1}[\nu_k < 0]$. Ground-state reaction intermediates require $n_{\text{imag}} = 0$, whereas transition states require strictly $n_{\text{imag}} = 1$.

The geometry used for the frequency calculation is compared with the optimized stationary-point geometry. For coordinate matrices $\mathbf{X}$ and $\mathbf{Y}$ centered at the origin, the geometry match requires $\delta_\pi(\mathbf{X}, \mathbf{Y}) = \max_i \|\mathbf{Y}_{\pi(i)} \mathbf{Q}_\pi - \mathbf{X}_i\| \leq 3 \times 10^{-6}$ Å, evaluated across element-conserving permutation mappings π and optimal Kabsch rotations $\mathbf{Q}_\pi$.

For a first-order saddle point with imaginary normal-mode displacement vector $\mathbf{u}$, the mode-alignment diagnostic compares the largest relative-displacement magnitude among monitored reaction-center pairs W with the largest among all atom pairs:

$$\eta = \frac{\max_{(i,j) \in W} \|\mathbf{u}_i - \mathbf{u}_j\|}{\max_{i<j} \|\mathbf{u}_i - \mathbf{u}_j\|}. \tag{2.11}$$

The diagnostic $\eta \in [0, 1]$ measures reaction-center participation relative to the largest pair displacement, using a comparison ratio of 0.5. It can identify spectator motion such as methyl wagging or solvent libration. The resulting alignment flag is advisory; scientific validation uses the first-order-saddle requirement and the same-TS bidirectional IRC and endpoint optimization/frequency chain.

2.3.3 IRC Direction and Endpoint Consistency

Intrinsic reaction coordinate (IRC) trajectories evolve from the saddle point along the steepest-descent path in mass-weighted coordinates $\mathbf{q} = \mathbf{M}^{1/2}\mathbf{x}$ [1, 2]:

$$\begin{aligned} \frac{\mathrm{d}\mathbf{q}_\pm}{\mathrm{d}s} &= -\frac{\nabla_{\mathbf{q}} U}{\|\nabla_{\mathbf{q}} U\|}, \\ \mathbf{q}_{\text{init},\pm} &\approx \mathbf{q}_{\text{TS}} \pm \epsilon \mathbf{v}_-. \end{aligned} \tag{2.12}$$

Cloudra checks that both IRC directions start from the verified saddle point and extracts the terminal frame from each direction. For paths between endpoint minima, each terminal frame is used to start an unconstrained geometry optimization. The resulting stationary point must have zero imaginary frequencies ($n_{\text{imag}} = 0$), and its canonical isomeric SMILES must equal that of the declared reactant or product, including topology and stereochemistry.

For elementary steps yielding dissociated molecular components, each dissociated product must independently converge to a verified minimum. Over the monitored pair set W, let $\mathbf{q}_{\text{TS}}$, $\mathbf{q}_{\text{E}}$, and $\mathbf{q}_{\text{R}}$ denote distance vectors at the transition state, terminal IRC frame, and reactant reference. With covalent cutoffs $c_{ij} = 1.25(r_i + r_j)$ and tolerance $\tau = 0.10$ Å, define $W_{\text{c}} = \{(i,j) \in W : d_{ij}^{\text{R}} \leq c_{ij}\}$ and interval $[a_p, b_p] = [\min(d_p^{\text{TS}}, d_p^{\text{R}}) - \tau, \max(d_p^{\text{TS}}, d_p^{\text{R}}) + \tau]$. The Boolean predicate is_between is satisfied when $\|\mathbf{q}_{\text{E}} - \mathbf{q}_{\text{R}}\|_2 < \|\mathbf{q}_{\text{TS}} - \mathbf{q}_{\text{R}}\|_2$ and $a_p \leq d_p^{\text{E}} \leq b_p$ for all $p \in W_{\text{c}}$. Dissociated-product paths must satisfy both the component-minimum checks and is_between.

2.3.4 Evidence Aggregation and Energies

For elementary step e, let K_e list its required calculation stages. Set $P_{e,k} = 1$ when the calculation results assigned to stage k satisfy its physical criteria, and $P_{e,k} = 0$ otherwise. All required stages must pass:

$$A_e = \bigwedge_{k \in K_e} P_{e,k}. \tag{2.13}$$

When explicit property checks R are declared, the step must also satisfy $A_e \wedge S_e$. Each stage is recorded as supported, pending, skipped, or rejected. A documented exemption or method limitation is thus recorded separately from a completed calculation.

The electronic activation barrier is computed directly from verified stationary-point energies:

$$\Delta E_e^{\ddagger} = E_{\text{TS},e} - E_{\text{ref},e}. \tag{2.14}$$

Energies share a consistent quantum chemical level of theory. For thermochemical profiles, zero-point energy, enthalpy, and Gibbs free-energy corrections are evaluated from harmonic partition functions under declared temperature and pressure specifications.

2.4 Calculation Diagnostics and Retries

The system distinguishes execution errors, numerical nonconvergence, and physical evidence problems. Execution errors include invalid input syntax, environment faults, and parsing failures. Numerical checks record self-consistent field (SCF) and geometry optimization convergence. Physical evidence problems include saddles with $n_{\text{imag}} \neq 1$, IRC endpoints awaiting verification, or molecular graphs corresponding to a different species. Imaginary-mode participation and geometric interval measurements indicate which atomic motions or distances to inspect.

Let T contain the species and elementary steps requested by the task. Extending $P_{e,k}$ to each $t \in T$, the calculation-stage requirements to be resolved are

$$U = \{(t,k) : t \in T,\, k \in K_t,\, P_{t,k} = 0\}. \tag{2.15}$$

For each $(t,k) \in U$, diagnostic feedback identifies the affected species or step and the problem: SCF nonconvergence, normal-mode misalignment, a missing IRC-to-endpoint calculation link, or a molecular graph mismatch. After a failed calculation, the system retains its output, updates U, and reports the geometry and convergence information to the agent.

Calculations that depend on a failed stage are rescheduled; earlier attempts and raw outputs are retained. The system reorders atom indices, assigns each calculation result to its species or step, and updates the graph status. The model revises the chemical constraints, dihedral angles, or electronic-structure method for the next attempt.

2.5 Implementation Defaults

The implementation described here corresponds to Cloudra revision 70daa5a.[18] Numerical arrays and linear algebra use NumPy[19]; quantum-chemical quantities are extracted with cclib alongside engine-specific native-log readers[20]; and benchmark graph representation and matching use NetworkX[21]. Preorganization uses 32 rigid orientations, up to 8 guided placements, minimum interfragment separation 1.0 Å, fragment-contact multiplier 1.7, and heavy-atom RMSD clustering threshold 0.5 Å. MMFF94 relaxation uses up to 2000 iterations per phase with a distance restraint force constant of 100.0. Standard IDPP employs 10 images optimized via MDMin (100 steps, force tolerance $f_{\max} = 0.1$, NEB spring constant $k = 0.1$); lightweight IDPP uses Adam optimization (4000 steps, step size 0.02, gradient tolerance 10^{-7}). Path convergence criteria evaluate the maximum force residual on intermediate path images.

The mapping coefficients and the $J_{\min} + 2$ ambiguity window are fixed ranking heuristics. Charge and radical mismatches carry larger costs than hydrogen-count and bond-edit mismatches. Branch crowding uses the fixed decay $\rho = 0.5$, traversal through depth five, and congestion flag $B_{\text{axis}} \geq 9.0$. The mode-participation diagnostic uses the comparison ratio 0.5 as an advisory flag.

Cloudra also provides a conformer-search tool with RDKit and external CREST backends.[22] The default preorganization described above uses rigid placements, guided MMFF94 relaxation, and clustering. CREST can be requested for additional conformer or noncovalent-complex sampling, followed by quantum-chemical optimization and physical validation.

2.6 CloudraBench construction and experimental design

2.6.1 Task coverage and tier rationale

CloudraBench evaluates end-to-end mechanism calculations on 50 tasks: 20 Easy tasks examine elementary transformations, conformational changes, and basic execution reliability; 10 Normal tasks examine intermediate identification and multistep continuity; 10 Hard tasks examine competing pathways and stereochemical identity; and 10 Impossible tasks examine the limits of the allowed methods and justified negative conclusions (Figure 5). Related reactions can appear in different tiers when they require different pathways, configurations, or electronic states. Tasks were selected using literature sources and reference calculations, considering both their ability to distinguish workflow performance and their computational cost. An accompanying catalogue lists each task's name, input conditions, and allowed methods.

2.6.2 Reference answers and evaluation protocol

Each task fixes the chemical conditions, charge, spin, deliverables, and allowed methods. Easy, Normal, and Hard tasks permit xTB[23], PM7[24], and B3LYP-D3(BJ)/def2-SVP[25–30]; Impossible tasks are restricted to xTB and PM7. Computational evidence is read from native ORCA, Gaussian, or MOPAC logs.[31–34] Reference answers specify species, elementary steps, stereochemical identities, and applicable numerical or negative targets. Numerical barriers are electronic energy differences at a consistent level of theory; stable-minimum references and actual IRC connectivity are verified separately. For accessible reaction steps requiring verification, TS frequencies, bidirectional IRCs, outer-frame provenance, and endpoint optimization/frequency calculations form the evidence chain. Species-property and electronic-state questions are checked against their specified targets.

2.6.3 Scoring and study configurations

Graph structure, self-consistency, and accuracy are evaluated separately and combined with weights of 0.1, 0.3, and 0.6. Graded physical support and runtime delivery completeness are reported separately. A correctly identified method limitation is reported as a supported negative conclusion. Reference-evidence completeness, special treatment, and qualifications awaiting resolution are documented per task; the presence of a reference graph and complete scientific validation are assessed separately.

2.6.4 Repeated-task study, primary comparison, and workflow ablations

The repeated-task study (P1b) compares Guided, Standard, Record, and Brief-errors on E1, NB1, HA1, and IA1, with three independent runs per combination (48 task-level runs). The repeated runs used DeepSeek-v4.1-flash (1017).[37] E1, NB1, HA1, and IA1 concern elimination preorganization, multistep decarboxylation, stereochemical ring opening, and electronic-structure method limits, respectively; their archived trajectories supply the subsequent local diagnostic cases. Its elapsed-time unit is a four-task batch at concurrency two, with three batches per configuration. The full-benchmark study (P2) retains one run for each of the 50 tasks and four configurations (200 task-level runs). All P2 configurations used DeepSeek-v4.1-flash (1017), called with the archived API alias `deepseek-v4-flash` through the official DeepSeek endpoint and streaming OpenAI-compatible chat completions, with requested minimal reasoning effort. Frozen configurations and actual request/completion records agree on this model identifier. The primary comparison is Guided against Standard using this same backbone; Standard is labelled Backbone in the figures. P1b and P2 are the archived identifiers of the repeated-task and full-benchmark studies. The P2 execution versions and request settings are listed in Supplementary Information Section S1.

Standard hides explicit DAG tools, removes enforcement of physical checks, and supplies neutral computational feedback. Guided retains the research graph, scientific checks, and detailed diagnostics. Record preserves available tools and records with optional physical validation; Brief-errors retains guidance while shortening diagnostics. Common task statements and method restrictions apply across configurations; run versions, budgets, stopping reasons, and artifact provenance are retained in the archive.

Before scoring, each run's existing research records are converted into a common mechanism graph. Human curators use the agent's stated mechanistic intent and conclusions to recover species and steps from its conversations, tool records, reports, and calculation files. They restore file references and record the source of each graph field. The evaluator searches the same run's sandbox for native calculation evidence and applies common scoring rules. Organization uses only calculations and scientific conclusions already present in the run; Gold supplies the evaluation targets. The final dataset contains all 200 runs and reports Graph,

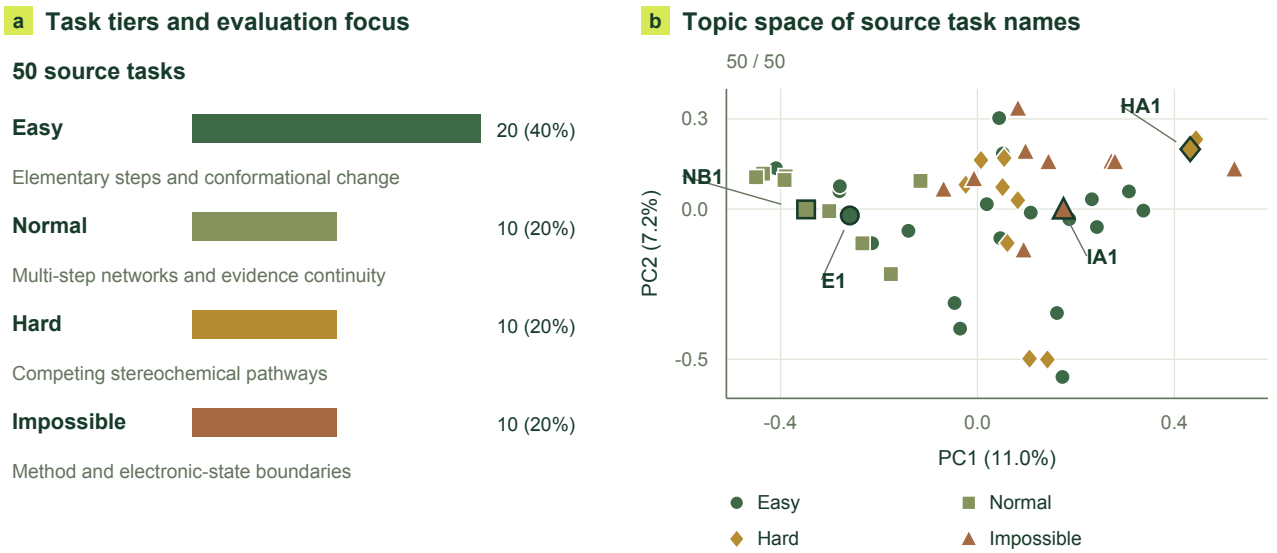


Figure 5. CloudraBench task composition and research themes. (a) Counts and evaluation focus of the four task tiers. (b) Two-dimensional PCA, computed with scikit-learn[35], of BAAI/bge-m3 embeddings[36] of all 50 original task names. E1, NB1, HA1, and IA1 are the four tasks used in the repeated-run experiment. Colors and marker shapes encode the predefined task tiers.

Self-consistency, and Accuracy alongside each composite score. Paired differences and 95% percentile intervals use 100,000 task-cluster bootstrap samples, retaining the four configurations of each task together (seed 20261031). Computational effort is measured on the 15 selected tasks with records for all four configurations, using quantum-chemical job counts and cumulative job-hours. These measurements are reported separately from the pilot's batch elapsed times.

3 Results and Discussion

3.1 End-to-end benchmark evaluation

CloudraBench measures end-to-end mechanism-calculation capability: constructing candidate mechanisms, carrying out the required quantum-chemical calculations, and reaching task-consistent positive or negative conclusions. We report composite scores together with their graph, self-consistency, and accuracy components. Offline organization assembles each mechanism graph from its run's own research materials and links it to the run's existing score records. The resulting 200-run table contains all three components for every run, with an overall composite mean of 0.7051. Figures 6–10 report runtime graph production, TS–IRC–endpoint evidence, and agreement with the task's answer.

3.1.1 Cloudra improves end-to-end mechanism-calculation scores with the same backbone

Guided obtained a mean composite score of 0.7930, compared with 0.5814 for Standard. Across the 50 task pairs, the mean improvement was 0.2116, with a 95% task-bootstrap interval of [0.1290, 0.2985]. Guided scored higher on 36 tasks, lower on seven, and equally on seven; the median paired difference was 0.1142 (Figure 6).

Guided's mean over its ten lowest-scoring tasks was 0.3221, compared with 0.0930 for Standard.

The most consistent difference occurred in the Hard stereochemistry tasks. Guided averaged 0.9577 and Standard 0.6888, a paired increase of 0.2689 [0.1213, 0.4459], with Guided scoring higher on all ten tasks. These tasks require assigning each barrier and computed pathway to the corresponding stereoisomer or competing channel.

On Normal tasks, Guided averaged 0.7370 versus 0.5731, an increase of 0.1640 [0.0365, 0.2922], with seven wins and three losses. Easy-task means were 0.9125 and 0.6989, respectively; the paired median difference was 0.0500, showing a different pattern from the broadly positive Hard comparison. Impossible-task means were 0.4451 and 0.2473. Their interpretation depends on whether the run addressed the requested method-limited negative target, so these tasks are examined separately below.

3.1.2 Self-consistency contributes most to the composite-score gain

The three components score different parts of the calculation (Figure 7). Graph measures the delivered coverage and identities of the requested species and steps. Self-consistency measures the support linking saddle points, frequencies, IRC starts, and endpoint optimization/frequency calculations. Accuracy measures agreement with the task's numerical or negative target. A run can therefore give a close numerical answer while leaving its supporting path incomplete, or establish a coherent local calculation that answers a different research question.

Across all 50 Guided–Standard task pairs, the weighted graph contribution to the mean improvement was +0.0108, self-consistency contributed +0.1179, and accuracy +0.0829, accounting together for the composite gain of 0.2116. For the 20 Normal and Hard tasks, the mean gain was 0.2164, with contributions of +0.0080 from graph, +0.1247 from self-consistency, and +0.0837 from accuracy. Within the ten Normal tasks, the corresponding contributions were +0.0098, +0.1141, and +0.0400; within the ten Hard tasks they were +0.0063, +0.1353, and +0.1274.

On Hard tasks, the weighted accuracy improvement is comparable in size to the self-consistency improvement.

HH1 provides a direct example. Guided and Standard both reported a binaphthyl racemization barrier of 23.48 kcal/mol, close to the 23.01 kcal/mol reference. Both had

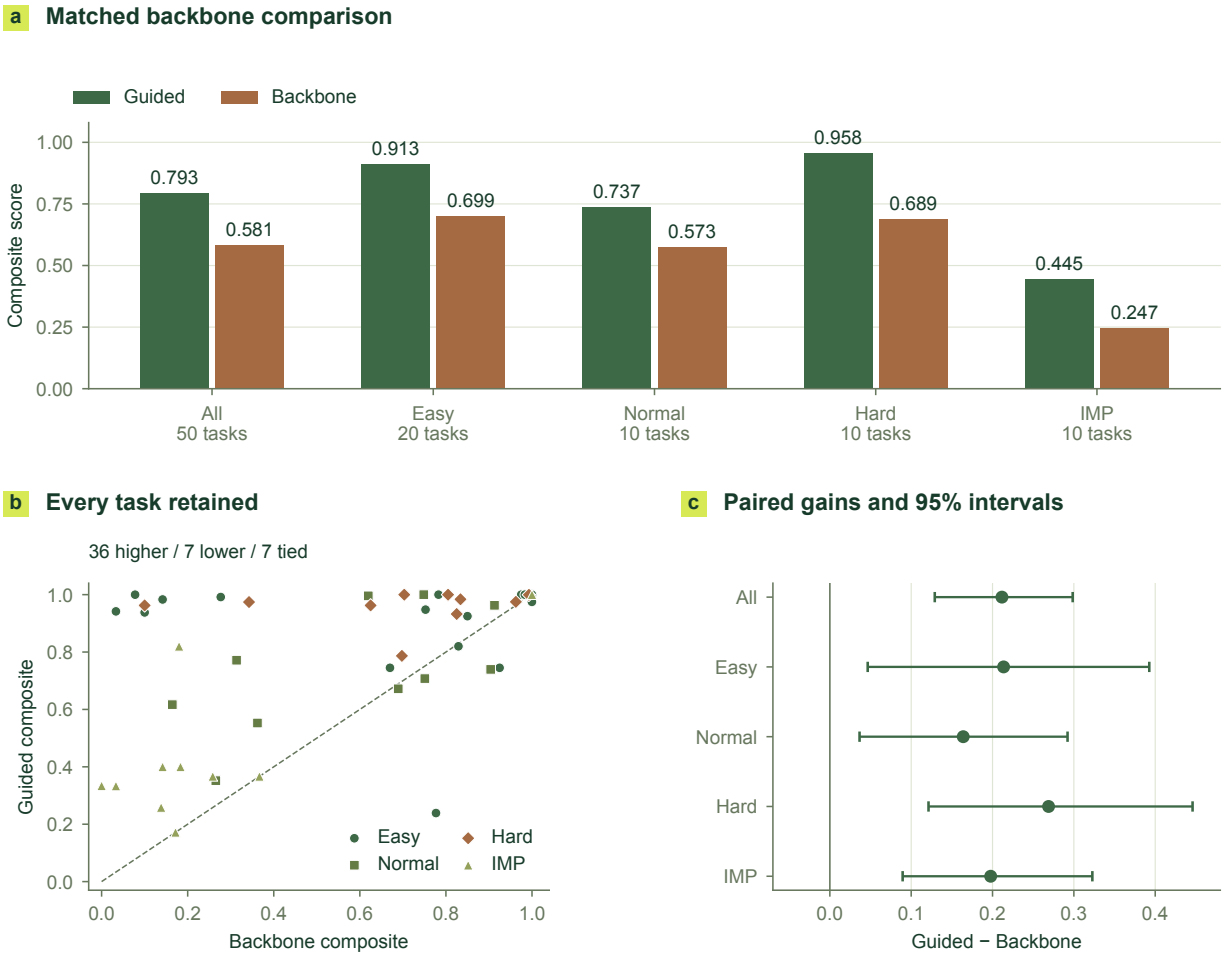


Figure 6. P2 benchmark outcomes under the archive-adjudicated track. (a) Guided and the same-backbone Standard baseline across all 50 tasks and each tier. (b) All 50 paired task outcomes. (c) Mean paired score differences with 95% percentile intervals from 100,000 task-level bootstrap resamples. Each task has one run per configuration; Easy contains 20 tasks and the other tiers contain 10 each.

graph scores of 1 and accuracy scores of approximately 0.973. Their self-consistency scores were 1.0 and 0.5, producing composite scores of 0.9838 and 0.8338, respectively. The archived Standard study contained candidate channels and IRC inputs; the required endpoint connections remained unverified. Guided attached the path evidence to the corresponding saddle and endpoints. HI1 shows a related pattern: accuracy was approximately 1 in both runs, while self-consistency increased from 0.2333 to 1.0 and the composite score from 0.7032 to 0.9999. Establishing which minima the saddle connected therefore required the additional IRC and endpoint calculations.

3.1.3 Delivery artifacts and graded scientific support remain distinct

Runtime root-graph production is recorded separately from the physical evidence found in each run (Figure 8). During the original runs, Standard delivered root graphs for 43 of 50 tasks, while Guided, Record, and Brief-errors each delivered 50. The seven Standard tasks requiring offline root-graph reconstruction were F1, HA1, HF1, IA9, M17, NR1, and P3. Offline organization assembled their graphs from the corresponding runs' existing materials, producing a complete 50-task set for every configuration. Recovering these seven Standard records credits research already performed by the baseline, making the comparison more conservative than withholding credit solely because the root delivery file was missing.

The seven Standard studies averaged 0.1537 after organization, compared with 0.8285 for Guided on the same tasks, and contributed 44.7% of the total Guided–Standard difference. Their individual scores ranged from 0.0333 to 0.3425. Calculations and candidate descriptions could remain in a run's records while the requested pathway or comparative conclusion was unfinished. Offline organization recovers that existing content for evaluation.

For the 40 non-Impossible tasks in each configuration, Guided had self-consistency exactly equal to 1 on 24 tasks (60.0%) and partial positive credit on 16. Standard had seven full-score (17.5%), 29 partial-score, and four zero-score tasks. Record had 16 full-score (40.0%) and 24 partial-score tasks; Brief-errors had 20 full-score (50.0%) and 20 partial-score tasks. Figure 8 presents these complete evidence-score distributions alongside the separate runtime graph-delivery counts.

3.1.4 Guided outperforms Record with the same available tools

Record averaged 0.6918 and Brief-errors 0.7541 across all 50 tasks. Guided exceeded Record by 0.1012 [0.0396, 0.1695], with 28 wins, seven losses, and 15 ties. On all 20 Normal and Hard tasks together, the mean Guided–Record

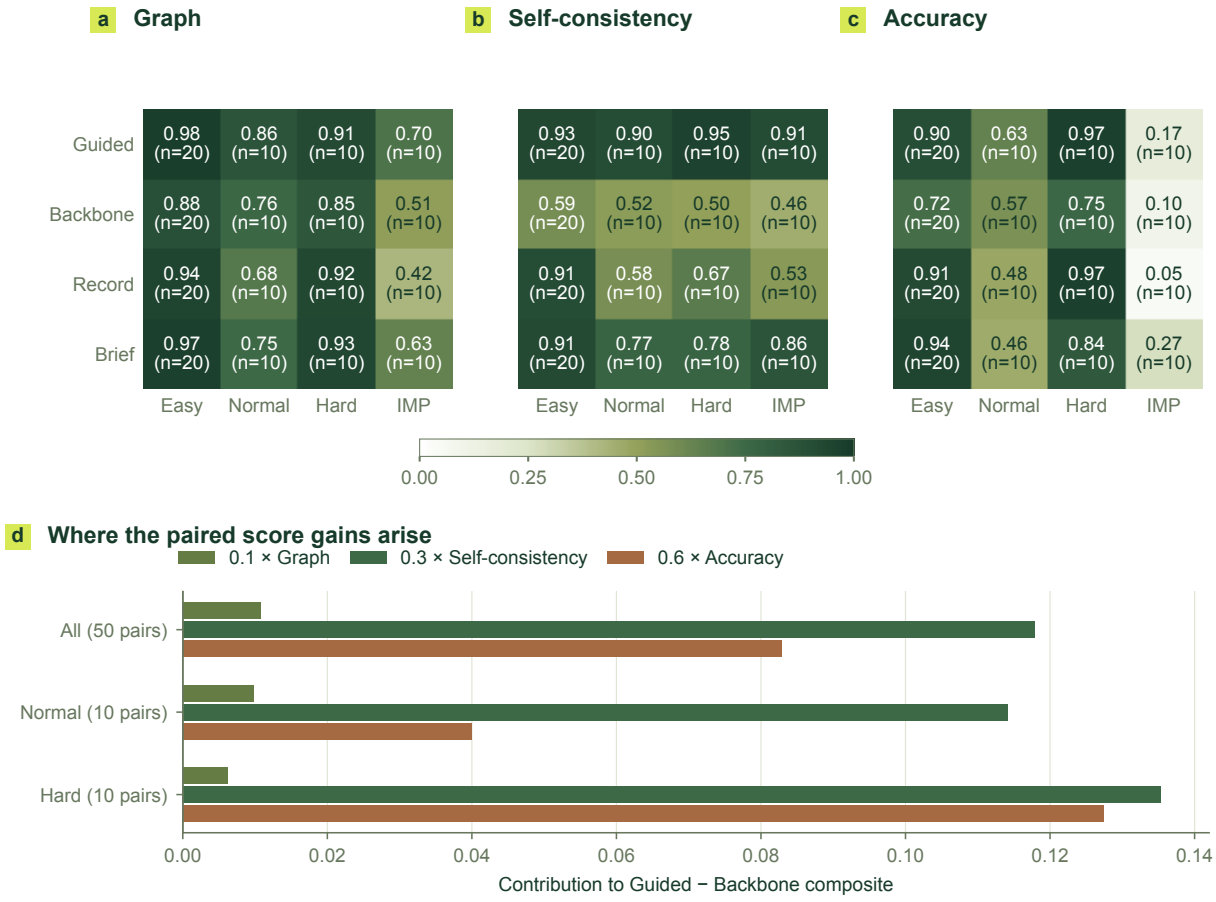


Figure 7. CloudraBench score layers across all 200 organized runs. (a–c) Means of Graph, Self-consistency, and Accuracy in each arm and tier, using 50 tasks per arm overall, 20 Easy tasks, and ten tasks in each other tier. (d) Guided–Backbone mean differences decomposed with the fixed weights 0.1/0.3/0.6, using all 50 pairs, ten Normal pairs, and ten Hard pairs. Components are aligned with the final organized totals using each run's recorded research materials and score records.

difference was 0.1468: graph contributed +0.0083, self-consistency +0.0919, and accuracy +0.0466. The largest contribution came from self-consistency, which scores the frequency, IRC, and endpoint calculations supporting the reported steps.

3.1.5 Research batch times and quantum-chemical job-hours

Figure 9 presents the timing information from the repeated-task study alongside the computational effort available for P2. For the repeated study, the measurement unit is a complete four-task batch at a concurrency of two, with three observed batches per configuration. Mean batch elapsed times were 4.451, 3.958, 3.894, and 4.827 hours for Guided, Standard, Record, and Brief-errors, respectively. Guided batches took 12.5% longer than Standard and 7.8% less time than Brief-errors on average. The figure shows all 12 observed batches, retaining their variation and original outcomes. These measurements include the batch's research lifecycle, waiting for calculations, and continuation after results return.

P2 effort is summarized over the 15 deeply inspected tasks shared by all four configurations. Guided used 11.08 job-hours per task, Standard 7.55, Record 7.88, and Brief-errors 15.59. The corresponding final composite means were 0.8124, 0.6495, 0.7360, and 0.7273. Guided used 46.7% more cumulative job time than Standard while increasing the mean score by 0.1629; relative to Brief-errors, it used 28.9% less job time and scored 0.0851 higher. Guided was both faster in cumulative job time and higher-scoring than Brief-errors on seven of the 15 paired tasks.

Across the common tasks, Guided submitted 789 recorded quantum-chemical jobs, Standard 419, Record 431, and Brief-errors 883. Recorded failed-job counts were 114, 45, 62, and 167, respectively, with cancellations recorded separately. The four difficult studies NU3, NU5, NU6, and NR1 accounted for 163.5 job-hours across their 16 runs, 24.3% of the 673.82 job-hours in the available 63-run effort archive. Some required steps in these four studies remained unresolved after additional attempts. In the archived Standard missing-graph cases, calculations could also outlive the research session, leaving subsequent results outside an active study.

Guided exceeded Brief-errors by 0.0388 overall, with an interval of $[-0.0317, 0.1076]$, 26 wins, ten losses, and 14 ties. The comparison varied by research demand: Guided's mean advantage was 0.1514 on Normal and 0.1295 on Hard, whereas Brief-errors had the higher Impossible mean.

3.1.6 Per-task comparisons of barriers, pathways, and calculation evidence

Figure 10 places all 50 tasks and four configurations alongside their complete score components. NB1 and NI1 also illustrate the scores obtained on multistep mechanisms. NB1

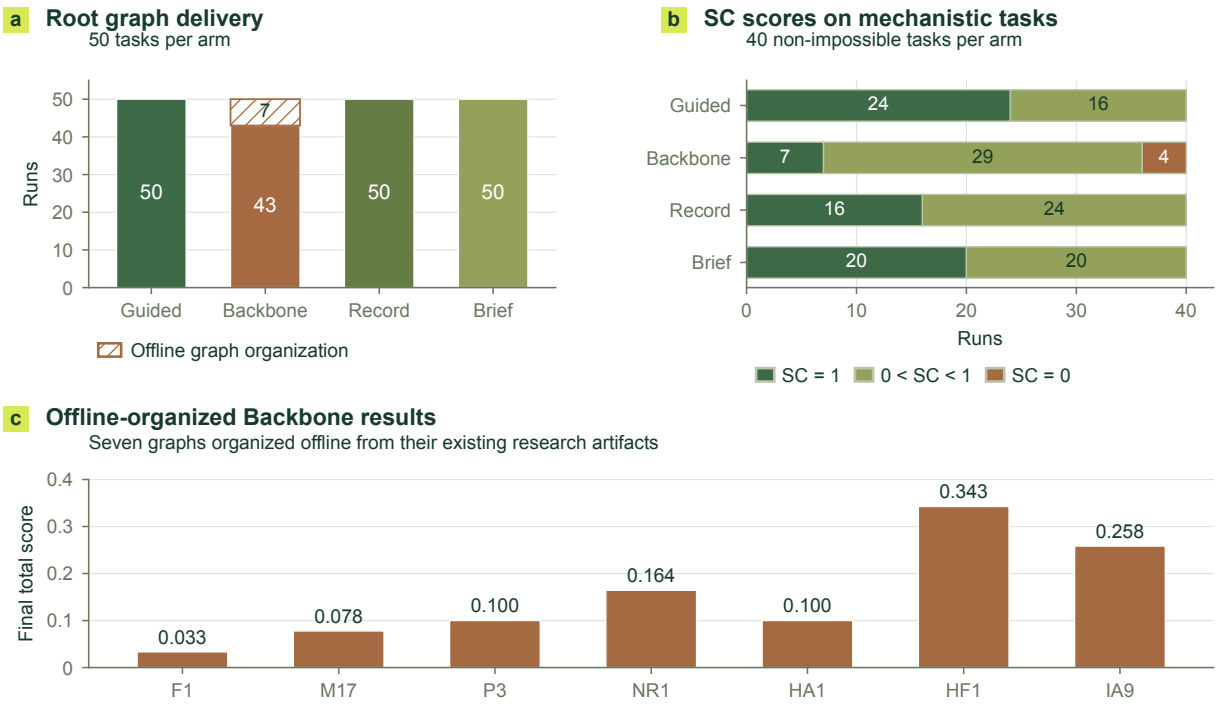


Figure 8. Runtime graph delivery and final scientific support in P2. (a) Original root mechanism graphs across 50 tasks per arm: Guided 50, Backbone 43, Record 50, and Brief 50. The hatched segment marks seven additional Backbone graphs assembled offline from the runs' own research materials; the organized set contains 50 graphs per arm. (b) Final continuous self-consistency (SC) scores for all 40 non-Impossible tasks per arm, separating SC = 1, partial positive credit, and SC = 0. Full-score counts are 24, 7, 16, and 20; Backbone has four zero-score tasks. (c) Organized composite scores for the seven Backbone tasks whose root graphs were reconstructed offline, ranging from 0.0333 to 0.3425 (mean 0.1537).

scored 0.9628 under Guided, 0.9128 under Standard, and 0.9900 under Brief-errors; Guided and Brief-errors both had self-consistency 1. NI1 scored 0.9958 under Guided and 0.6196 under Standard, with self-consistency 1 and 0.35. In contrast, NQ5 favored Record over Guided, and NU3 favored Brief-errors.

The archived records explain several barrier differences (Figure 11). In HA8, Guided reported 29.46 kcal/mol and Standard 20.43 kcal/mol against a reference of 29.48 kcal/mol. The archive traces this 9.03 kcal/mol difference to the energy reference: the lowest all-s-trans reactant minimum in Guided and a reactive cisoid conformer in Standard. The IRC identifies the local structures connected by the saddle; the reported activation energy uses the stable-minimum reference specified by the task.

NU6 exposes a related issue in path identity. Guided's recorded concerted saddle had an energy of -247.030042392 Ha and the betaine minimum -247.060060699 Ha, giving a difference of 18.836772 kcal/mol. Guided described this as an energy cost involving the concerted saddle region while retaining the target betaine-to-product ring-closure saddle as unresolved. Standard delivered 18.837 kcal/mol as the second-step ring-closure barrier. The two reports assign this nearly identical energy difference to different processes. Guided's attempts from interpolated betaine structures and constrained scan peaks repeatedly relaxed toward the direct reactant-to-product saddle region. The saddle's path identity therefore required another assessment from its optimized geometry and pathway.

In G2, Guided scored 0.2389 and Standard 0.7769. Guided's native logs supported an approximately 0.845-kcal/mol double-H-transfer barrier. After three rejections of numerical barrier declarations on a companion edge, the agent removed the numerical fields. Supervisor feedback prompted restoration; a fourth rejection and a subsequent saddle-evidence check preceded their final removal, together with the saddle energy/log bindings. The final graph retained the values in text and received accuracy 0. Brief-errors retained the corresponding numerical result and scored 0.8500.

The accuracy values of 0.7 in D1, D3, and L2 followed the specified cap on numerical-deviation penalties. D1's barrier difference from Standard traced to the reactant-conformer reference; D3's mainly reflected different saddle energies. L2 distinguished reactive cZc and lower tZt references. Standard delivered both associated barriers, 18.99 and 28.28 kcal/mol, and reference matching selected the former. The Supplementary Information provides the final workflow-level components, native evidence, and delivery records for these five tasks.

3.1.7 Impossible tasks distinguish local support from the requested negative conclusion

The Impossible tier tests whether research under xTB/PM7 reaches the specified method-limited conclusion. Brief-errors averaged 0.4829, Guided 0.4451, Standard 0.2473, and Record 0.2325 (Figure 12). Brief-errors exceeded Guided on three tasks and scored lower on five, with two ties. Its mean advantage of 0.0379 was concentrated in particular answers: IA1 contributed +0.8283 and IA9 +0.5666, whereas Guided's IG8 result contributed a difference of

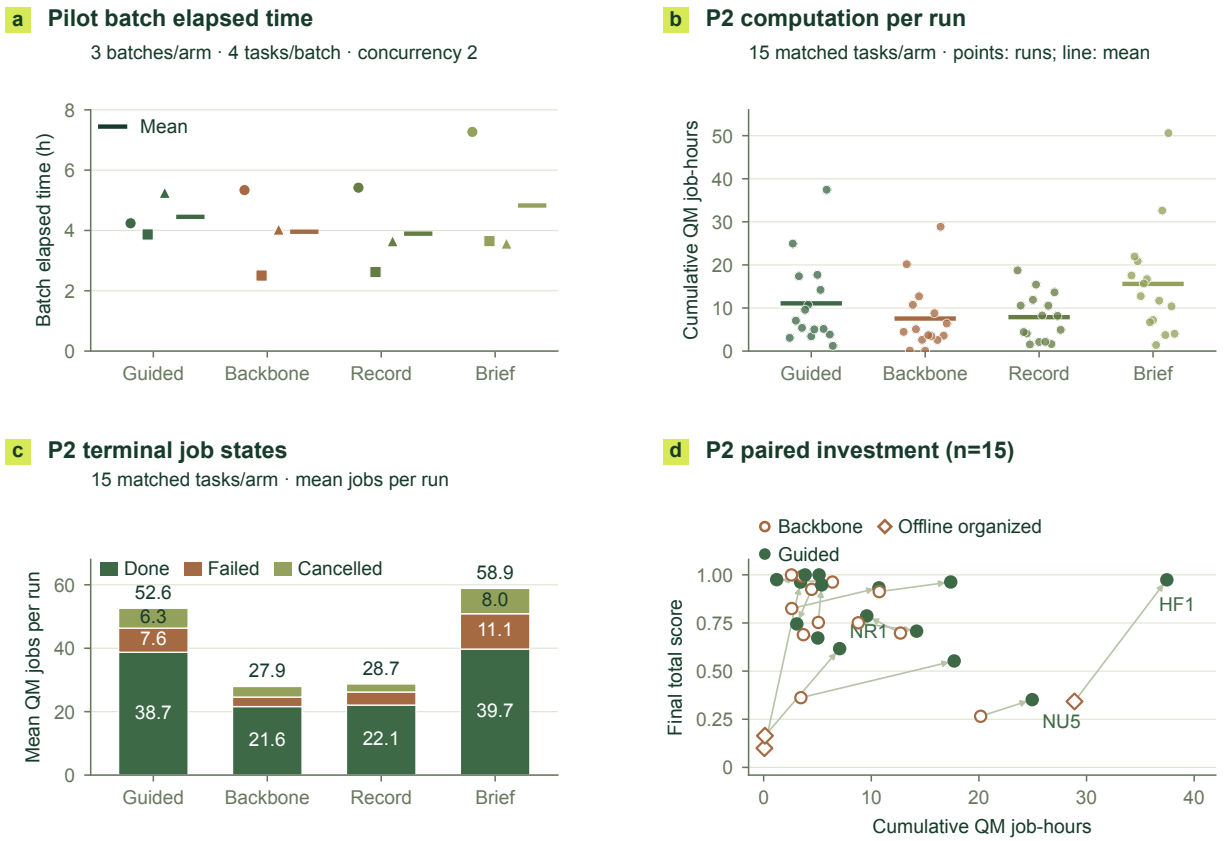


Figure 9. Research elapsed time and computational investment. (a) Twelve P1b four-task batches at concurrency two, three per arm; lines mark means. (b) Cumulative QM process durations for the 15 P2 tasks shared by all arms. (c) Mean completed, failed, and cancelled jobs in that cohort. (d) Same-task Backbone-to-Guided changes in investment and final score; diamonds mark offline-organized Backbone graphs. Job-hours sum process walltime; panel a measures batch latency. Backbone/M1 effort is reconstructed from native CPU time divided by four.

−0.6661 in the Brief-errors-minus-Guided direction.

In IA1, concerning cyclobutadiene's square-geometry automerization target, Brief-errors had graph, self-consistency, and accuracy scores of 1, while Guided's accuracy was 0. In IA9, concerning dioxetane thermolysis, Brief-errors and Guided both had self-consistency 1, with accuracy 1 and 0, respectively. IG8 showed the opposite accuracy ordering for the chromium-dimer target. Across the seven tasks with the same accuracy score in Brief-errors and Guided, Brief-errors averaged 0.0500 lower. Its overall Impossible advantage decomposed into weighted contributions of −0.0066 from graph, −0.0155 from self-consistency, and +0.0600 from accuracy. The higher mean arose from the requested conclusions on specific tasks, while Guided retained stronger local support on several others.

Across all 40 Impossible runs, four had accuracy 1 and 14 combined self-consistency 1 with accuracy 0. For example, both Guided and Brief-errors obtained self-consistency 1 and accuracy 0 in IA5, IC3, and IF3 for the requested negative target. Guided and Brief-errors both received partial accuracy of 0.7 in IB9, whose requested network contains positive and negative components.

The saved Brief-errors IC3 graph provides a concrete example of this distinction in scientific reporting. It marks the lowest open-shell singlet and its state-change edge inaccessible and withholds the requested singlet–triplet gap. The computed closed-shell singlet remains as a separate characterized branch, and the two method-dependent relative energies are explicitly excluded from the requested gap. The run therefore retained useful exploratory results while declining to promote them into an answer about a different electronic state. Its final accuracy score remained 0. The discrepancy between this explicit refusal and its accuracy score is examined in the Discussion.

3.2 Molecular identity, distance comparison, and fragment rotation

We next examine whether the model distinguishes molecular identities, compares distances between specified atoms, detects mismatched calculation references, and identifies missing IRC calculations. Archived research trajectories supply the molecular structures and calculation records. Controlled local studies with Qwen3.8:27b vary the supplied information, answer order, and selected internal states to test their effects on each judgment.

3.2.1 Equivalent SMILES and answer order affect molecular-identity judgments

A geometric modification first requires identifying the molecule, atoms, and reaction center to which it applies. A molecule can be represented by different SMILES traversals, while a molecular formula can describe different connectivities. To examine how the model handles these distinctions, we constructed 16 molecular-identity pairs covering identical identities, connectivity changes within the same molecular formula, changes in connectivity and charge location, double-bond stereochemistry, and protonation changes.

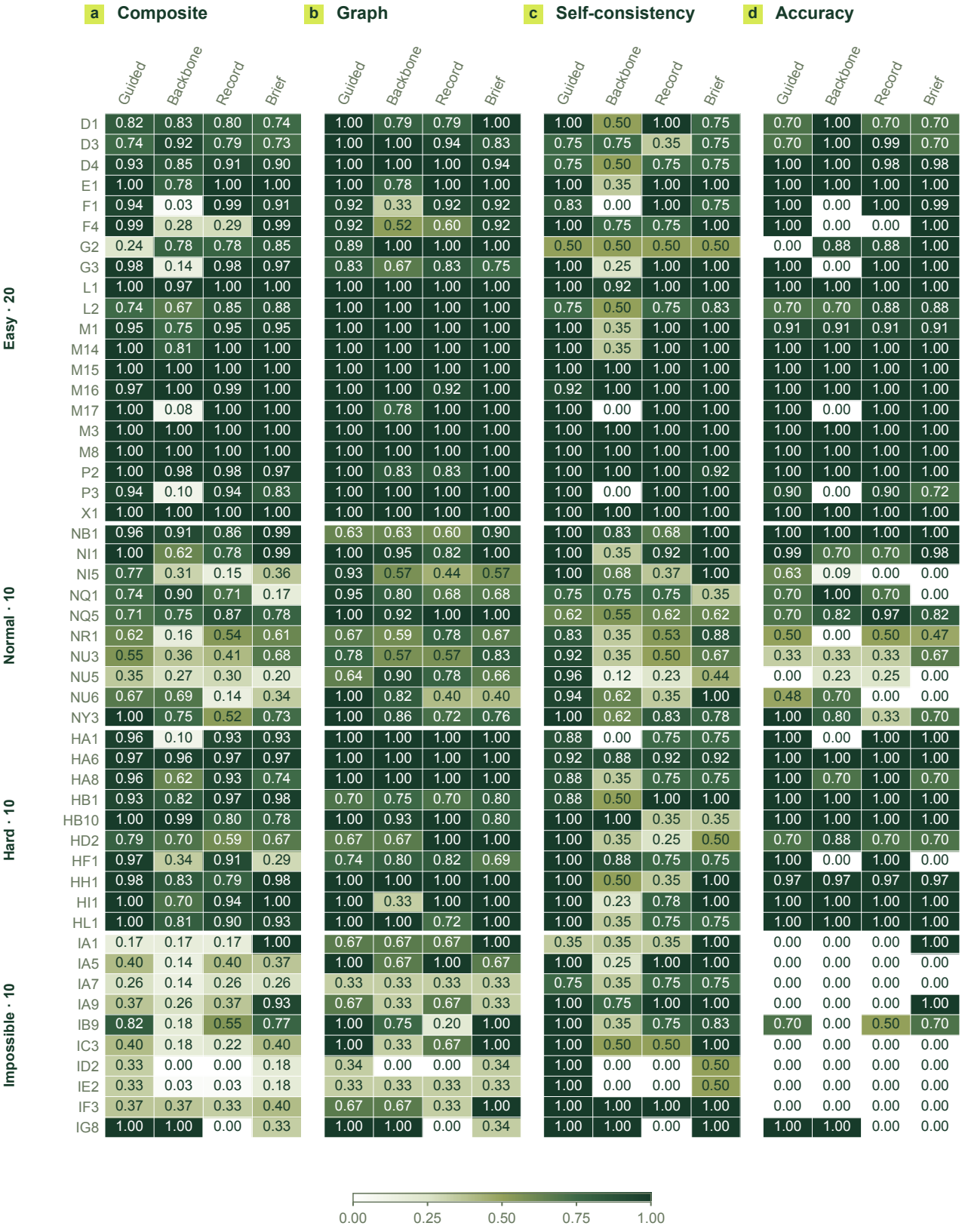


Figure 10. Complete CloudraBench results for all 50 tasks and four configurations. The aligned blocks show composite, Graph, Self-consistency, and Accuracy on the same 0–1 scale; all 800 cells are present. Values are rounded to two decimals; the source table retains full precision.

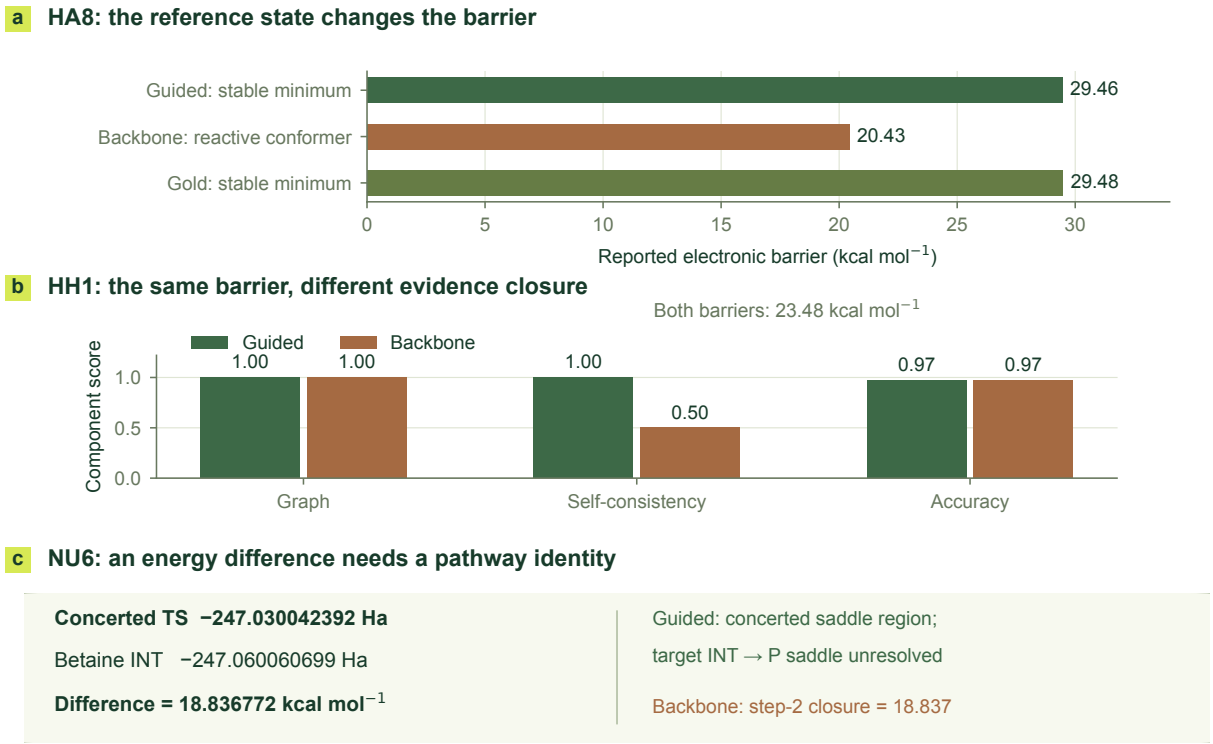


Figure 11. Energy references and pathway assignments in matched Guided–Backbone tasks. (a) HA8 barriers use a stable-minimum or reactive-conformer reference and are compared with campaign Gold. (b) Both HH1 runs report 23.48 kcal/mol with similar accuracy but different self-consistency scores. (c) NU6's recorded energy difference is 18.836772 kcal/mol: Guided reports the cost of reaching the concerted saddle region and leaves the target ring-closure saddle unresolved; Backbone reports 18.837 as the second-step ring-closure barrier. Values and assignments derive from the archived records.

Each pair was presented using the original strings or equivalent rewritings and with two answer orders, yielding 64 judgments.

The model answered 58 correctly. Equivalent rewriting changed the semantic judgment in 4 of 32 matched representation comparisons; reversing answer order changed the judgment in 6 of 32 matched option-order comparisons. All six of the latter cases involved repeated selection of the letter A. One C_5H_8 pair in Figure 13 illustrates this behavior directly: an endocyclic and an exocyclic double bond define different connectivities, yet the model answered A under all four conditions, although A meant "same" in one order and "different" in the other. Thus, changing the notation or answer order can change whether the model judges two structures to be the same molecule.

Cloudra stores molecular connectivity separately from coordinates and uses atom correspondence to identify which atoms to measure, move, or compare. Product-endpoint alignment, path interpolation, and reaction-center measurements retain this correspondence as the structure's coordinates change.

3.2.2 Geometric measurements reduce repeated numerical reasoning

We selected four actual structures from the NB1 trajectories and asked the model which of two specified interatomic distances was shorter. Each structure was presented with both answer orders and with either a complete XYZ geometry or program-measured distances. These two information conditions were further crossed with direct-answer and thinking modes, producing 32 native chat calls (Figure 14).

In direct-answer mode, all eight responses to XYZ inputs selected A, with four correct answers. Providing measured distances produced eight correct answers and preserved the semantic answer when option order was reversed. With thinking enabled, the XYZ condition produced seven correct answers; the remaining response reached the output limit of 1,024 generated tokens. Increasing the budget for that input alone yielded a correct answer within 1,353 generated tokens.

All eight answers in the measured-distance condition were correct. Across the eight paired comparisons with thinking enabled, the median generated token counts were 807.5 for XYZ inputs and 86 for measured distances.

Providing the distances removes the need for the model to calculate them from XYZ coordinates. Cloudra similarly computes bond lengths, bond angles, and dihedral angles in software and uses deterministic operators to rotate or translate the selected fragments. The model chooses the atoms and target geometry, then compares the measured distances or angles of the resulting candidates. Assigning these numerical operations to software follows the approach of program-aided language models.[38]

3.2.3 Feedback must match the specified atoms and geometric objective

In an actual E1 trajectory, the model sought to bring the dihedral angle defined by the transferring hydrogen H7, C2, C3, and leaving atom Cl4 close to 180°. It rotated the fragment on the C3 side about the C2–C3 axis by +120° and −120°, producing candidates A and B, and set the C3–Cl4 distance to 2.35 Å. The representative dihedral in the default structure report involved H5, whereas the current objective involved H7. The model then explicitly measured H7–C2–C3–Cl4 in



Figure 12. Impossible-tier outcomes and the brief-diagnostic ablation. (a–c) Composite, self-consistency (SC), and accuracy for all ten tasks and four arms, using the complete organized scores on a common 0–1 scale. Backbone denotes the actual same-backbone Standard arm. (d) Paired Brief-minus-Guided differences: the mean advantage of 0.03788 is concentrated in IA1 and IA9, with an opposite outcome in IG8; Guided scores higher on five tasks, Brief on three, and two tie. (e) The mean difference decomposes into weighted accuracy (+0.06000), Graph (-0.00661), and SC (-0.01550); the small discrepancy in their sum reflects rounding of saved totals. The paired task-bootstrap 95% interval is -0.188 to +0.284; task-specific claims are analyzed below.

both candidates, obtaining approximately 59.4° and 179.4°, and used candidate B in its next actual submission.

Figure 16 shows Newman projections along C2→C3 drawn directly from the saved candidate coordinates, and Figure 17 summarizes the corresponding target-angle residuals. Candidate B places the H7 target angle within 0.6° of 180°, compared with a residual of 120.6° for A.

The corresponding residuals for the H5 comparison were 124.2° and 115.8° in A and B, respectively. Identifying the anti-periplanar H7–Cl4 arrangement in B required the H7-based dihedral.

When local probes provided only H5 measurements, the model requested an additional H7 measurement under all three cyclic option permutations. Adding H7 measurements led it to select B under all three permutations. Two further paired tests of information sufficiency also shifted from requesting additional measurements to judging the information sufficient when the target quantity was added.

3.2.4 Activation replacement affects distance answers differently across option orders

To test how a reported distance affects the answer, we performed activation replacement at a single token position in the same local quantized model.[39] One distance in the input was changed from 2.029 Å to 1.029 Å, while the other remained at 1.248 Å, reversing the correct comparison answer. We then transferred the residual-stream state at the numeric token position from the original input into the modified input at layers 7, 15, 31, and 47, and read the full-vocabulary probability assigned to the answer supported by the original fact.

The effect varied with layer and option order (Figure 20). With reversed options, the probability of A, the original-fact answer, increased from 0.074 for the modified input to 0.956 and 0.718 after replacement at layers 7 and 15, respectively. Replacements at layers 31 and 47 yielded 0.072 and 0.070. With the original option order, the probability of answer B changed from 0.009 to 0.029 and 0.015 after replacement at the first two tested layers, while the last two remained close to the modified-input baseline.

Self-replacement controls and control replacements at an unchanged prefix position produced zero logit differences across the full vocabulary.

With reversed options, replacing the early-layer state restored the original-fact answer as the preferred choice; with the original option order, the modified-fact answer remained preferred. Recovering the answer from the same numeric position therefore depended on option order, as did the identity judgments in Section 3.2.1.

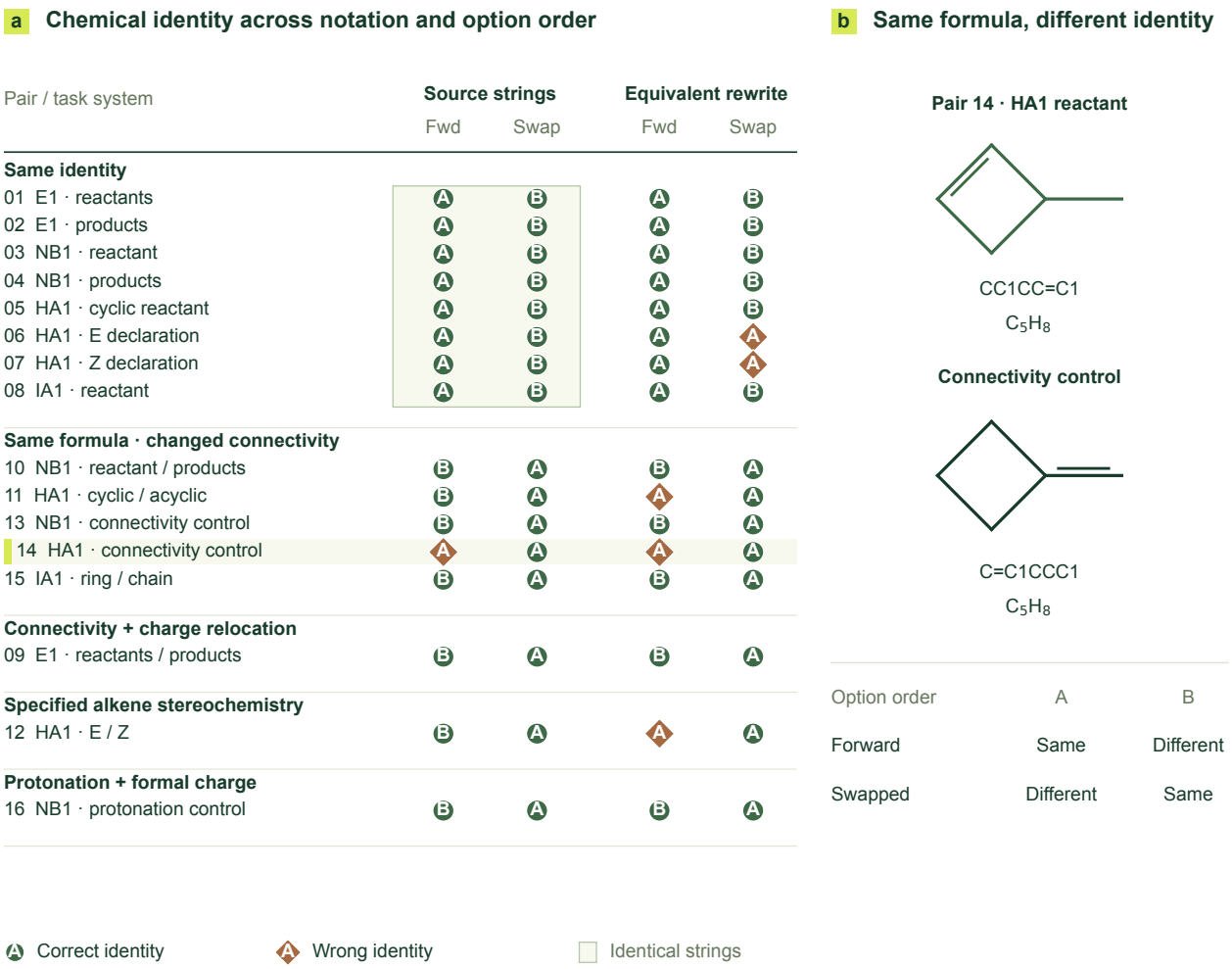


Figure 13. Molecular-identity judgments under changes in molecular representation and option order. Each of the 16 molecular pairs has four input conditions. Circles and diamonds indicate correct and incorrect judgments, respectively; the letters retain the model's actual answers. Shaded regions denote identical-string conditions. The right panel shows the pair-14 connectivities `CC1CC=C1` and `C=C1CCC1`, both C_5H_8, and the meanings of the answer options.

3.3 Identifying the species and pathways established by calculations

3.3.1 Associating frequencies and IRC endpoints with the same saddle

Once geometric objects enter computation, optimized structures, energies, frequencies, and paths must be assembled into support for the same reaction step. The local validity and attribution of each result jointly determine this support chain: frequencies must correspond to the declared saddle point, the IRC must start from that saddle point, and optimizations and frequencies for the final frames on both sides must correspond to the respective endpoints. The DAG attaches the optimized geometry, frequency calculation, IRC start, and endpoint calculations to the corresponding species or step. It records job execution, optimization convergence, and these physical checks separately; a converged optimization can still require frequency or endpoint calculations.

We extracted four source evidence chains from the archive and constructed versions that preserved the links or changed a single reference. Crossing these with flat and linked representations and two answer orders yielded 32 inputs. The altered references concerned the frequency geometry, IRC starting geometry, endpoint-optimization input, or target species; local termination, convergence, energy, and frequency values were held constant. Figure 22 illustrates the E1 case: changing the frequency geometry reference from the saddle point to an existing endpoint changes how that frequency result supports the current step.

All 16 judgments with linked representations were correct, compared with 14 of 16 with flat representations. Both errors occurred when a reference had been changed and the options reversed, in E1 and the second NB1 step. In subsequent native follow-up calls on two selected source chains, enabling thinking yielded correct judgments for all 16 judgments of the evidence relationships, including the two previously incorrect cases.

The model detected every altered reference in the linked format, even though the local numerical values were unchanged. This presentation effect is consistent with observations that graph encoding affects relational reasoning in language models.[40] Cloudra records the corresponding artifact references for each species, step, and calculation attempt, identifying which saddle, IRC, and endpoint calculations are being compared.

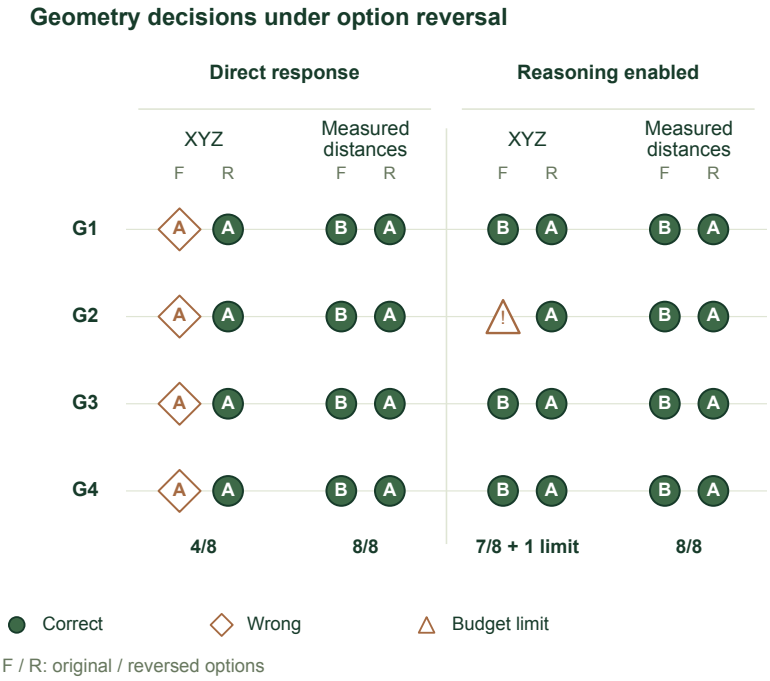


Figure 14. Judgments of which of two specified atom-pair distances is shorter, using coordinates or measured distances. Four source structures are compared under direct response and reasoning-enabled conditions, each with original (F) and reversed (R) option order. Circles indicate correct judgments, diamonds incorrect judgments, and the triangle a response that reached the output budget. Letters retain the actual answers.

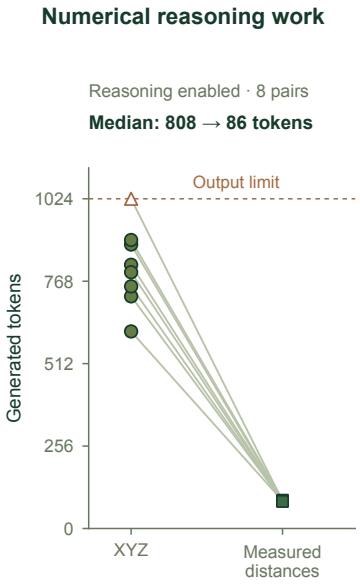


Figure 15. Generated-token counts with reasoning enabled. Each line connects Cartesian-coordinate input and measured-distance input for the same source structure and option order; eight paired comparisons are shown. The dashed line marks the output limit.

3.3.2 Diagnostic feedback on IRC connectivity and transition-state curvature

Diagnostic feedback connects an evidence defect to the chemical object and calculation that require revision. Cloudra's structured diagnostics identify specific problems in quantum-chemical outputs, including steric-clash atom pairs, imaginary-mode misalignment, and disconnected trajectory endpoints. The agent can then change the clashing geometry, inspect the imaginary mode, or recalculate the affected endpoint.

Two archived trajectories illustrate this connection. In a Brief-errors HA1 trial, compact error codes replaced the detailed IRC-endpoint linkage diagnostics. The Computational Agent repeatedly substituted source-minimum references while the geometric disconnection remained unresolved. In a Guided NB1 trial, the verification found zero imaginary frequencies and returned a normal-mode diagnostic. The Coordinator requested a retry with revised transition-state guess parameters and optimization constraints, which the Computational Agent used in the rescheduled calculation.[41]

3.3.3 IA1: a completion request with a missing IRC calculation

An archived IA1 trajectory shows how a completion request can conflict with the recorded scientific requirements. The reaction step had mandatory bidirectional short_irc verification pending, and the waiver list was empty. The Supervisor had earlier commented positively on intermediate optimization convergence. The model subsequently requested completion while the IRC requirement remained unresolved.

The harness rejected that request and returned the outstanding short_irc requirement to the model. In the next submission, the model changed its status to partial completion (Figure 24). The missing IRC calculation remained unresolved when the completion status changed.

We also tested whether the model could identify the outstanding IA1 requirements from the recorded state. All 12 judgments were correct (100%) across six pairs with reversed answer-option order, using both a minimal summary and an extended 12-message history.

3.4 Workflow guidance, scientific judgment, and outlook

Cloudra applies atom-mapped geometric operations to selected atoms and represents species and elementary steps in the Delivery DAG. For each step, the graph associates the saddle with its frequency calculation, IRC, and endpoint calculations. The 50-task comparison measures end-to-end mechanism-calculation performance with these methods. Guided had higher mean self-consistency and accuracy than Standard; in Hard tasks, their weighted contributions to the gain were +0.1353 and +0.1274. HH1 and HI1 report reference-consistent barriers; identifying the minima connected by those saddles still requires IRC and endpoint calculations.

HA8 and NU6 show two assignments that must be checked after geometry optimization. In HA8, the stable reactant minimum defines the barrier reference, while the IRC connects the saddle to a local reactive conformer. In NU6, an initial guess for betaine-to-product closure repeatedly optimized toward the direct reactant-to-product saddle region. The resulting saddle must therefore be assigned from its optimized structure and path. The DAG records the reactant conformer, saddle, IRC endpoints, and assigned step so that the agent can inspect which structures each energy belongs to after a new calculation. This use of recorded calculation results is consistent with the role of external planning and

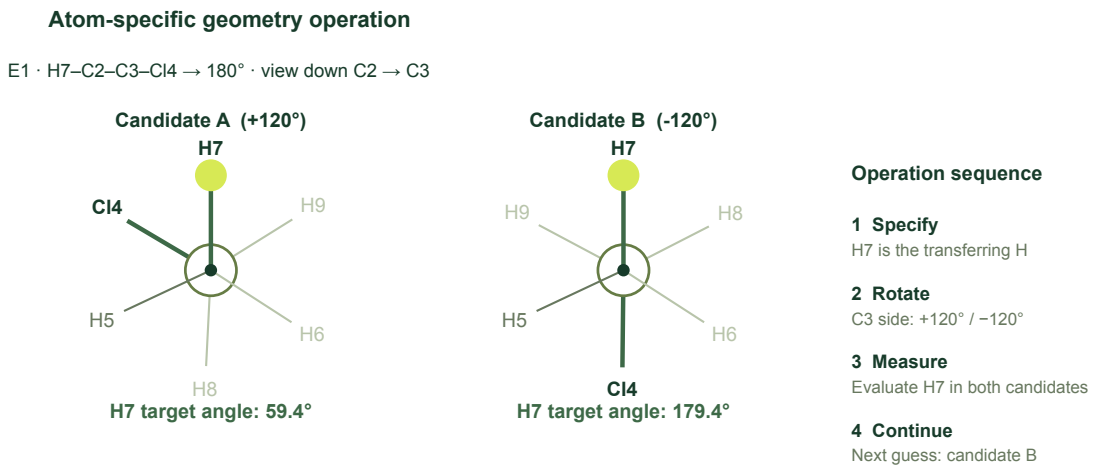


Figure 16. Target-specific geometry operations in E1. Newman projections retain the directions of the original candidate coordinates viewed down C2→C3. H7 is the specified transferring atom. Candidate A and candidate B give target angles of 59.4° and 179.4°, respectively. The operation sequence is shown alongside the projections.

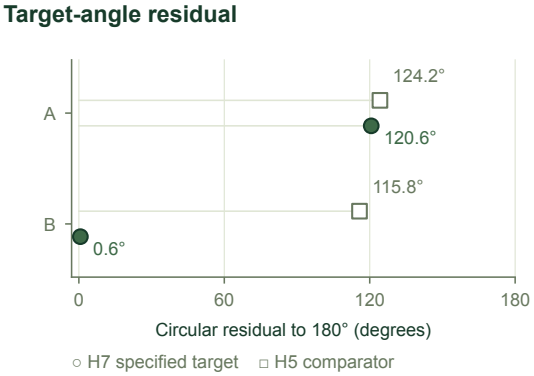


Figure 17. Circular angular residuals from the specified 180° target for candidates A and B. H7 is the specified target atom; H5 is the comparator.



Figure 18. Next-action choices under three local option permutations. When only H5 measurements are supplied, the model requests H7 measurements; with H7 supplied, it selects candidate B.

verification feedback in language-model workflows.[42]

A missing endpoint optimization calls for another endpoint calculation; a saddle that repeatedly returns to another route calls for a revised guess or pathway hypothesis. An electronic state beyond the method's scope requires reassessing the method and the requested conclusion. The agent must compare its reaction description with the computed structures and paths, because a plausible explanation can persist after the underlying result has changed.[43] Record retained the tools and obtained lower self-consistency than Guided, particularly on Normal and Hard tasks. The Brief-errors comparison also differed by task: detailed feedback accompanied higher Guided means on Normal and Hard, while Brief-errors had the higher Impossible mean. These results support examining whether feedback directs the agent to the relevant geometry, calculation, or method limitation, consistent with studies of tool feedback in reasoning correction.[41, 44]

Brief-errors/IC3 retained a characterized closed-shell branch while withholding the requested lowest open-shell singlet and its singlet–triplet gap. Brief-errors also scored higher on IA1 and IA9, whereas Guided scored higher on IG8. These crossed outcomes motivate a task-dependent hypothesis: detailed diagnostics may help repair a TS or IRC calculation, with limited benefit from further local repair when the method or target needs reconsideration. The present comparisons document final task outcomes; the causal effect of diagnostic length and its underlying internal mechanism remain questions for dedicated study.

In G2, checks on a companion edge blocked numerical declarations, after which the agent removed values already supported by native calculations. Supervisor feedback restored the fields; further rejection preceded their deletion from the final graph. While the pathway is being tested, the numerical observation and its native log should remain available, with the current validation status specifying which mechanistic conclusions the value supports. G2 retained partial self-consistency credit and lost accuracy credit when the numerical fields were removed.

The requested target and positive alternative shared endpoint connectivity in Record/IA5 and Brief-errors/IC3: disrotatory and conrotatory paths in the first case, and open-shell and closed-shell states in the second. In both cases, the archived matcher selected the alternative positive object despite the run's explicit rejection of the requested target. The manuscript retains the final scores and reports these claims alongside them. Evaluating these tasks requires matching the reaction channel or electronic state before scoring the final conclusion.

A conditional target-assignment analysis in the Supple-

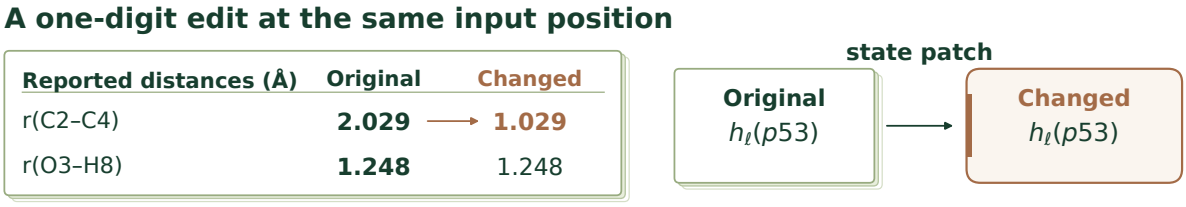


Figure 19. Intervention on a single reported distance. A one-digit edit changes the C2–C4 distance from 2.029 to 1.029 Å while O3–H8 remains 1.248 Å. The original-input residual state at token position 53 is inserted into the changed-input run at a specified layer. Token positions are zero-based.

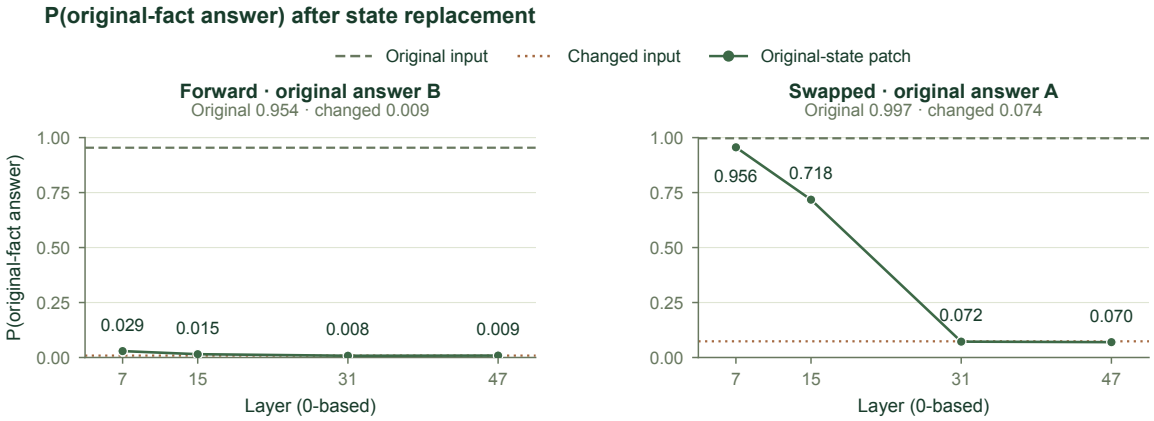


Figure 20. Original-fact answer probability after state replacement. Curves show full-vocabulary probability mass for the original-fact answer under forward (B) and swapped (A) option order, including bare and leading-space token spellings. Horizontal lines show original-input and changed-input baselines. Layer indices are zero-based.

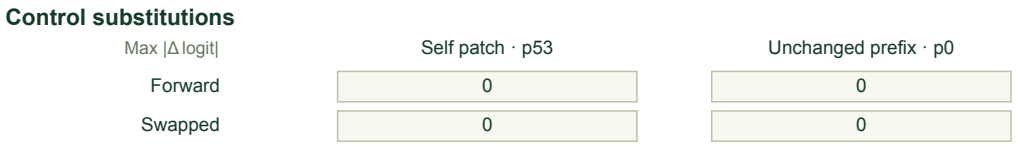


Figure 21. Control substitutions for the state-intervention experiment. Same-state self replacement at p53 and replacement of the unchanged prefix at p0 preserve all vocabulary logits in both option orders. The table reports the maximum absolute logit difference. Token positions are zero-based.

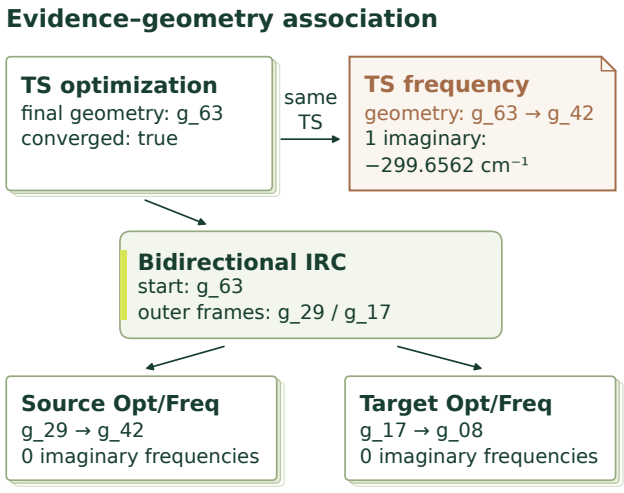


Figure 22. Changing the geometry assigned to the E1 frequency calculation. The frequency geometry reference changes from g_63 to g_42 while local termination, convergence, energy, and frequency values remain fixed. Arrows connect the TS, IRC, and endpoint optimization/frequency calculations.

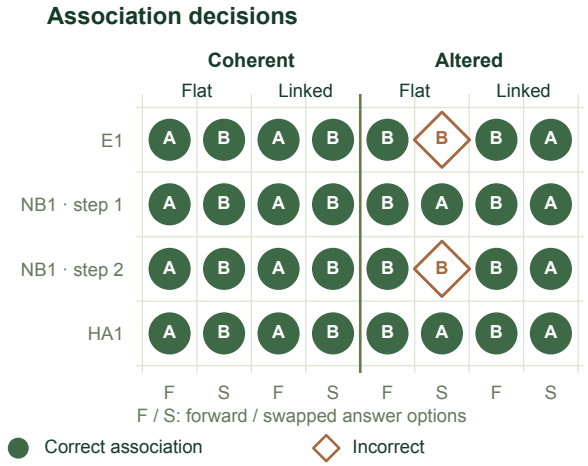


Figure 23. Judgments of whether calculation references support the specified step. The 32 inputs cross four source chains, coherent/altered links, flat/linked presentation, and forward/swapped answer order. Saved A/B responses and correctness markers are retained.

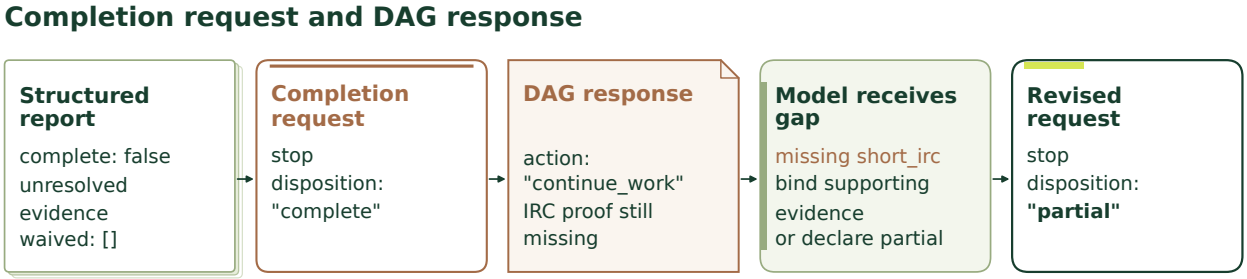


Figure 24. Completion requests and DAG feedback in the original IA1 trajectory. A report with unresolved evidence is followed by a completion request, a response identifying missing short_irc proof, feedback to the model, and a revised partial-completion request.

mentary Information applies the existing negative-credit rule to the intended IA5 and IC3 objects while holding their Graph and Self-consistency values and the other 198 runs fixed. Each assignment changes one composite from 0.4 to 1.0, increasing its workflow's Impossible mean by 0.0600. With both assignments, Brief, Guided, Record, and Standard have conditional means of 0.54294, 0.44506, 0.29250, and 0.24733. The ordering among Brief, Guided, and Record is unchanged; Record moves above Standard. Every Guided–Standard pair remains unchanged. The manuscript reports the final evaluation scores; these conditional values change only which graph object receives the existing negative-target credit.

P2 contains one retained run per task and configuration. Its task-cluster bootstrap intervals describe variation across the sampled task mix; same-task stochastic variation requires independent repeats. The separate P1b study supplies three repeats on each of four tasks under each configuration; these 48 runs cover four of the 50 P2 tasks. The complete organized P2 table covers all 200 runs and their three score components. Some Hard-task accuracy targets concern qualitative channel identity, so the reported gain includes pathway identification and evidence completion as well as numerical barriers. Repeating the full task set would characterize run-to-run variability and the frequency with which each research route is selected.

The Impossible tier concerns targets judged physically nonexistent under the task's permitted methods; these judgments are specific to that method description. Its component values also use negative-target rules: self-consistency 1 on an explicit negative conclusion is distinct from a positive TS–IRC–endpoint chain. In the 14 Impossible cases with self-consistency 1 and accuracy 0, the score must be read alongside the particular state, pathway, or property that the run characterized or rejected.

The 12 pilot batches measure real elapsed time for four tasks at concurrency two, including the original failures and stopping outcomes. The P2 effort archive covers selected deep-dive studies, with 15 tasks shared across configurations; cumulative job-hours represent recorded quantum-chemical process durations, with a documented native-log reconstruction for Standard/M1. They are distinct from core-hours, financial cost, and end-to-end P2 research latency. Concurrent calculations shared computational resources, and CPU queueing may have delayed individual results. Separating queueing, calculation, and agent-active time would show how much of each run was spent waiting, computing, or deciding what to do next.

Standard's missing graphs followed premature completion while calculations were still pending; the Guided/HI1 session remained waiting after the completion notification. In both cases, a completed calculation needed to resume the unfinished study before the final report was assembled. A further search problem is repeated optimization toward the same stationary-point region, as in NU6. Candidate-selection methods could compare new saddles with earlier attempts and help the agent choose the next guess or pathway hypothesis.

Cloudra's default preorganization samples 32 rigid relative placements, applies local MMFF94 relaxation, and selects candidates by energy and clustering. Frequency verification characterizes the candidate stationary point; a search for lower conformers requires additional sampling. HA8 requires both the specified stable reference and the local reactive conformer; a global-minimum claim additionally requires an assessment of conformer coverage. CREST is available through a separate optional backend; its coverage across the 200 runs remains to be established. Highly flexible systems may require broader sampling and more quantum-chemical calculations. Further work should measure which low-energy conformers are recovered and how calculations are distributed among competing candidates.

The benchmark scores electronic energy differences. Zero-point, thermal/entropic, standard-state, and solvation contributions belong to a further free-energy treatment. This scope matters for NI5, specified in weakly acidic aqueous solution, and NU5, specified in dichloromethane at high concentration. Their scores assess the prescribed electronic-energy references. For association and dissociation, entropy and concentration-dependent standard-state terms enter $\Delta G^{\ddagger}$; solvent participation can also change the chemical model required to describe a pathway. Inferring solution-phase rate control or stereoselectivity would require a consistent free-energy treatment of the relevant species and pathways. The present electronic-energy comparisons leave the magnitude and direction of the solvent effect for a separate solvent-sensitivity study.

The atom-mapping coefficients and ambiguity window, and the decay, depth, and congestion settings in the branch-crowding descriptor, were fixed across the workflow comparisons. The study evaluates these fixed settings together; their sensitivity and individual contributions remain to be measured. The imaginary-mode alignment ratio of 0.5 is advisory; the stationary-point and IRC/endpoint evidence determine scientific validation. All P2 configurations used DeepSeek-v4.1-flash (1017), with the API alias

deepseek-v4-flash and the recorded request settings. Comparisons with additional backbones would test how the workflow gains carry across models.

Extending Cloudra beyond small-molecule mechanism calculations will require representations, computational methods, and physical criteria appropriate to the new systems. Its existing computational engines and shared execution infrastructure can support those calculations. Cloudra will be released as open-source software under the GNU Affero General Public License (AGPL), supporting inspection, reuse, and community development of the system.

Data availability

The retained P2 benchmark records, the 48-run P1b study, Gold reference-calculation evidence, and supplementary reproduction materials are publicly available at https://drive.google.com/drive/folders/1yZo3eJuCUuAgedQztgCtLyJcc8MZipf7. Package contents, versions, and checksums are described in Supplementary Information Section S7 and the accompanying download records.

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